



High frequency factor analysis with partially observable factors

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ABSTRACT

This paper considers a novel factor structure – *Partially Observable Factor Model* – where both observable factors and latent factors exist in the model simultaneously. Such factor structure can make sure both interpretability and goodness-of-fit at the same time. Necessary estimation methodologies for this partially observable factor model are developed in this paper for the high frequency data. The proposed estimation methodology is robust to jumps, microstructure noise and asynchronous observation times simultaneously.

When the observable factors are exogenous, we provide the estimation theory for the integrated eigenvalues of the residual covariance matrix, which including the bias-corrected estimator, central limit theorem and asymptotic variance estimator. As a result, the asymptotic normality of the bias-corrected estimator can be applied to test the existence of the latent factors.

When the observable factors are endogenous, we propose a novel framework of high frequency unsupervised exogenous component learning (HF-UECL), which can help people quantify the contributions of the observable factors into the latent factors. This is the first work on high frequency instrumental variables, and it can be regard as a necessary and non-trivial extension of the Projected-PCA in the world of continuous-time model. Statistical inferences have been established for the loadings of the observable factors onto the latent factors.

Monte Carlo simulation demonstrates the validity of our estimation methodologies. Empirical study demonstrates that (i) in the exogenous setting, the latent factors significantly exist in the residual process of the high frequency regression; (ii) in the endogenous setting, the correlations between the observable factors and latent factors do exist significantly.

1. Introduction

Factor model (or, factor analysis) has wide applications in statistical science, e.g., asset pricing models; disaggregate business cycle analysis; monitoring, forecasting, and diffusion indices; consumer theory, etc. In the recent years, the high frequency factor analysis received more and more attention for three reasons. First, high frequency data has had a rapid increase in amount during the last decade, which can overcome the curse of dimensionality in the low frequency factor analysis. Second, high frequency data can be observed in many fields of modern science, including chemical physics, molecular biology, material science, earth science, medical research, signal processing, finance and econometrics, see, e.g., Pavliotis and Stuart (2007), Boninsegna et al. (2018), Lee et al. (2017), Damouras et al. (2010), Calderon et al. (2008, 2009, 2013), Mykland and Zhang (2012), etc. Third, as the theoretical

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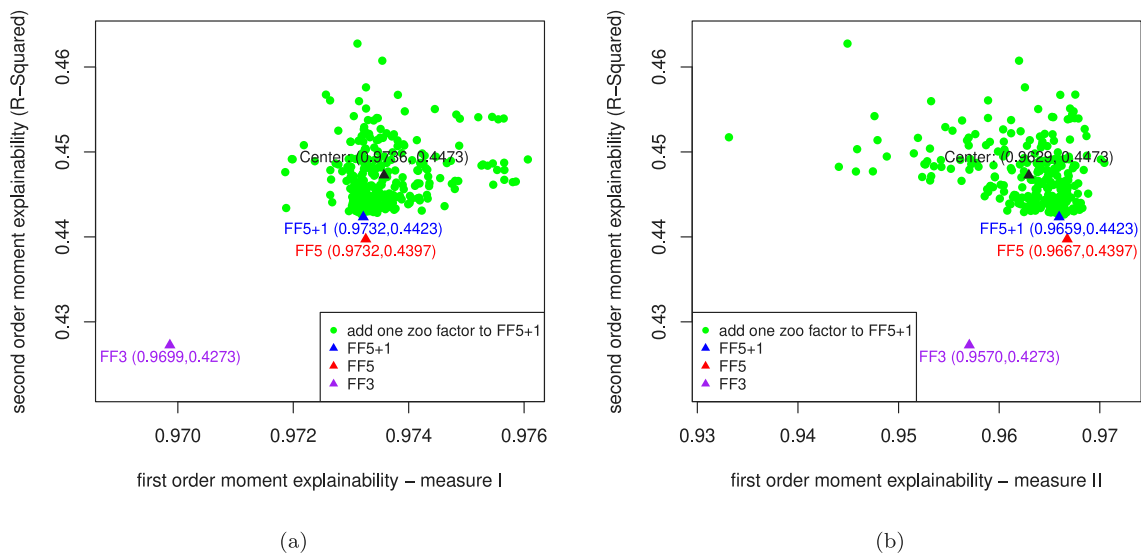


Fig. 1.1. Explainability comparison in first-order moment and second-order moment.

foundation of high frequency data analysis, the continuous-time modeling approach can relax the stationary assumptions in the traditional low frequency factor analysis and can allow for the time-varying features in the factor loadings and the covariance structure of the factors.

In the existing literature about the estimation of factor model, the common factors are assumed to be either fully latent/unobservable; or fully observable. When the common factors are fully latent/unobservable, principal component analysis (PCA) is usually applied to estimate the factor model, see, e.g., Chamberlain and Rothschild (1983), Stock and Watson (1998, 2002), Bai and Ng (2002, 2006), Bai (2003), Fan et al. (2013), and Giglio and Xiu (2021) for low frequency data; Aït-Sahalia and Xiu (2017, 2019), Pelger (2019, 2020), Kong (2017), Bollerslev et al. (2019), Dai et al. (2019), Chen et al. (2020), and Kong et al. (2021) for high frequency data. When the common factors are fully observable, regression is usually applied to estimate the factor loadings, see, e.g., Lehmann and Modest (1988), Fama and French (1992, 2017), Fan et al. (2011) for low frequency data; Fan et al. (2016a), Mykland and Zhang (2006), Aït-Sahalia et al. (2020) and Chen et al. (2023) for high frequency data.

Factor model has two main usages: (i) explaining the cross-sectional variations in expected excess returns (i.e., first order moment); (ii) explaining the cross-sectional variations in variance/covariances (i.e., second order moment). Either the purely observable factor model or the purely latent factor model are not perfect in general purpose. For example, although the purely observable factor model has good economic interpretations and satisfactory explainability in first order moment, its second order moment explainability is relatively low. This fact is evident based on Fig. 1.1, where the measures for first-order moment explainability are defined as: (I) $1 - \frac{1}{N} \sum_{i=1}^N \frac{|\hat{\alpha}_i|}{\frac{1}{T} \sum_{t=1}^T |r_{i,t}|}$; (II) $1 - \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \frac{|r_{i,t} - \hat{r}_{i,t}|}{|r_{i,t}|}$, where $\hat{\alpha}_i$ is the alpha estimate by incorporating the Fama–French 5+1 Factors. In contrast, the purely latent factor model can explain the second-order moment well but lacks good economic interpretations.

Besides the purely observable or latent factor model, this paper considers a novel factor model, where (i) only part of the common factors are observable, (ii) the rest of the common factors are assumed to be latent/unobservable. In other words, both observable factors and latent factors exist in this factor structure simultaneously. This is the first time that such factor structure is considered in the econometric literature. Here, we take the example of asset pricing model to illustrate the rationality of such factor model. In asset pricing theory, there are several observable risk factors which can explain the cross-sectional variation of the asset prices, e.g., market factor, size factor, style factor, momentum factor, etc. These observable risk factors are well-established both empirically and theoretically: (i) the profitabilities of these observable factors are evident in the real financial data; (ii) the reasons of their profitabilities are also well-documented in the finance/economic theory or behavior science. The high frequency version of these risk factors is also well implemented: (i) the high frequency factors constructed in Aït-Sahalia et al. (2020); (ii) the ETFs which serving as the proxies of risk factors, see, e.g., Size/Style ETFs and Sector ETFs. As shown in Fig. 1.1, the imbalanced explainability of the observable factors in first-order moment and second-order moment motivates us to consider the possible latent factor structure in the residual process of the high frequency regression. One might argue that if we consider all of the common factors are latent, we can apply PCA to estimate the factors, which can explain as much total variation as possible. However, the first several principle components generated in PCA do not have very clear economic interpretations, which is the limitation of such methodology. Therefore, if we put the observable factors and latent factors together into the factor model, and assume that the observable factors are orthogonal to the latent factors, we can reach the two goals at the same time: (i) econometricians can obtain as much economic interpretations as possible in the partially observable factor model; (ii) the partially observable factor model can explain more cross-sectional variation in the asset prices.

In this paper, we mainly develop the estimation methodology for the partially observable factor model under the high frequency setting, where the data generating process is described by the continuous-time models. For the cross-sectional dimension, the estimation methodologies are developed for both exogenous regressors and endogenous regression.

Under the settings of exogenous regressors, i.e., the observable factors and latent factors are orthogonal, we take a two-phase estimation method for the partially observable factor model. As the first step, we apply the high frequency regression technique for the observable factors, which generates the estimates of factor loadings and residual processes. In the second step, to estimate the latent factors, we apply the high frequency PCA technique to the residual processes obtained from the high frequency regression in the first step. Under this general framework, the estimates of the latent factors are the first several principal components generated from the residual processes and their covariance matrix. Moreover, the eigenvalues of the residual covariance matrix become the variances of the latent factor estimates, i.e., residual principal components (residual PCs). We construct the bias-corrected estimators for the eigenvalues of the residual covariance matrix. The central limit theorems are established for the bias-corrected estimators and related asymptotic variance estimators are also proposed, see, e.g., Section 5. As shown in the mathematical proof, the derivation of these theoretical results is completely new and non-trivial. With the well-established asymptotic normality of the bias-corrected estimator, econometricians can directly construct the hypothesis test for the variance of each latent factor estimate, of which the null hypothesis is that the variance equals zeros. The rejection of the null hypothesis indicates the existence of the latent common factor.

Under the settings of endogenous regressors, i.e., the observable factors and latent factors are correlated, we propose the high frequency unsupervised exogenous component learning method (HF-UECL), which is the first endogenous learning method in high frequency econometrics. This endogenous learning methodology is “unsupervised” in the sense that the exogenous component in this scenario is unobservable. In mathematical essence, our paper applies a singular value decomposition (SVD) to determine the loadings of the observable factors. As comparison, in the low frequency settings, the endogenous regression analysis with unobservable IVs are carried out by Projected PCA proposed by Fan et al. (2016b); Instrumented PCA proposed by Kelly et al. (2020); and Economic-Target PCA proposed by Bryzgalova et al. (2023). Given the solution of observable factor loadings, the projection of the latent factors on the direction of observable factors (hereafter, latent factor projection) can be derived accordingly, which provides us an important indicator about the co-movement level among observable factors and unobservable factors. In the theoretical results, we have developed the feasible estimators for the observable factor loading and the latent factor projection, and corresponding central limit theorems are also established for them.

The accuracy of high frequency factor analysis relies on the robustness of the covariance estimator. In fact, the typical features of high frequency data, such as jumps, market microstructure noise, and asynchronous/irregular observation times, can introduce different types of biases into the covariance estimator, see, e.g., Chen (2024). Such biases can greatly contaminate the factor structure of the intraday asset prices because the eigenvalue estimates and eigenvector estimates of the covariance matrix are also biased if the covariance estimator is not robust to the features of high frequency data. For example, the factor structure of the cross-sectional microstructure noise can be mistakenly regard as the latent factors in the intraday asset prices if the covariance estimator is not robust to market microstructure noise. Therefore, in order to get rid of the possible biases and to detect the true latent factor in the partially observable factor model, we apply the Truncated S-TSRV proposed in Chen (2024) to estimate the covariances, which is robust to jumps, market microstructure noise and asynchronous/irregular observation times simultaneously.

The statistical inference in this paper rooted in the estimation theories of integrated volatility functional, see, e.g., Jacod and Rosenbaum (2013) for the functionals with polynomial growth, which was later extended to the functionals under spatial localization assumption by Li et al. (2017) with a uniform convergence strategy. The relevant literature include but not limited to Li and Xiu (2016), Chen (2018), Li et al. (2019), Ait-Sahalia and Xiu (2019), Li et al. (2021), Chen et al. (2020, 2023), Chen (2024). Since the dependent structure in microstructure noise has been observed in recent years, see, e.g., Jacod et al. (2009) and Li and Linton (2022), there are some novel literature considering more stylized features in microstructure noise, which include but not limited to Kim et al. (2016) and Oh et al. (2024). Moreover, although we assume there are fixed number of observable factors throughout this paper, there are also some literature considering the inference about high dimensional and high frequency regression, see, e.g. Chen et al. (2024), Shin and Kim (2023, 2024), Kim et al. (2024).

The practical implication of this paper is that we provide the necessary adjustment for the estimation methodologies of high dimensional covariance/precision matrices which rely on purely observable factor models. Due to the limited explainability of the observable factors in second-order moment, the well-known low rank + sparse structure becomes infeasible in this situation. In contrast, the proposed partially observable factor model and the corresponding adjusted theories in estimating the high dimensional covariance matrices (see, e.g., Appendix C) can address the problem. For example, as documented in Section 8.3, the proposed partially observable factor model outperforms the purely observable factor model in constructing the minimum variance portfolio.

Empirical study is implemented for the intraday returns of the constituents of S&P 100 Index. The observable factors used in this study are the ETFs serving as the proxies of market factor and size/style factors. The estimates and confidence intervals of the eigenvalues in the residual covariance matrix are computed for each trading day within a 11-year time horizon (between January 2007 and December 2017). Under the exogenous assumption, the data analysis shows that the variance of the first residual principal component (PC) is always significantly greater than zero within the 11-year time window. The variances of the rest of the residual PCs are insignificant over time. Under the endogenous assumption, we find that there are around 20% of the latent factor projection estimates are significant across the 11-year window. Both of these observations from real financial data provides the evidence about the existence of latent factors in the partially observable factor model. We have also conducted the anomaly tests for the newly generated residual PCs and unsupervised exogenous components. In this study, the new anomalies have been identified.

For a matrix $\mathbf{M}_{p \times q}$, $\mathbf{M}^{(k, \cdot)}$ denotes its k -th row, $\mathbf{M}^{(\cdot, r)}$ denotes its r -th column, $\mathbf{M}^{(r, k)}$ denotes its (r, k) -th element, $d\mathbf{M}_t = \left\{ d\mathbf{M}_t^{(r, k)} \right\}_{1 \leq r \leq p, 1 \leq k \leq q}$, $\mathbf{M}^\top$ denotes its transpose, and $\mathbf{M}^+$ denotes its Moore–Penrose inverse in the sense of Magnus and Neudecker (2019, Section 2.5, pp.36). We denote the largest and smallest eigenvalue of matrix $\mathbf{M}$ by $\lambda_{\max}(\mathbf{M})$ and $\lambda_{\min}(\mathbf{M})$, respectively. $\text{tr}(\mathbf{M})$ denotes the trace of matrix $\mathbf{M}_{p \times p}$, i.e., $\text{tr}(\mathbf{M}) = \sum_{i=1}^p \mathbf{M}^{(i, i)}$. We denote by $\|\mathbf{M}\|$, $\|\mathbf{M}\|_1$, $\|\mathbf{M}\|_F$, $\|\mathbf{M}\|_{\max}$ the spectral norm, L_1 -norm, Frobenius norm and elementwise max norm of matrix $\mathbf{M}$, defined as $\|\mathbf{M}\| = \lambda_{\max}^{1/2}(\mathbf{M}^\top \mathbf{M})$, $\|\mathbf{M}\|_1 = \max_j \sum_i |\mathbf{M}^{(i, j)}|$, $\|\mathbf{M}\|_F = \text{tr}^{1/2}(\mathbf{M}^\top \mathbf{M})$, $\|\mathbf{M}\|_{\max} = \max_{i, j} |\mathbf{M}^{(i, j)}|$. If $\mathbf{M}$ is a vector, then $\|\mathbf{M}\|$ and $\|\mathbf{M}\|_F$ are equal to its Euclidean norm. For two sequences, we write $x_n \asymp y_n$ if $x_n = O_p(y_n)$ and $y_n = O_p(x_n)$. Let S_d denote the Euclidean space of all $d \times d$ real-valued symmetric matrices. S_d^+ is the subset of S_d , which contains all positive-semidefinite matrices. S_d^{++} denotes the subset of S_d^+ which contains all positive-definite matrices. For any $\mathcal{X} \in S_d^+$, $\lambda(\mathcal{X}) = (\lambda^{(1)}(\mathcal{X}), \lambda^{(2)}(\mathcal{X}), \dots, \lambda^{(d)}(\mathcal{X}))^\top$ is the vector of its eigenvalues in a nonincreasing order.

The article is organized as follows. Section 2 presents the high frequency factor model with partially observable factors. Section 3 provides detailed assumptions of data and process of interest. Section 4 presents the estimator of spot covariance. Sections 5 and 6 shows the details of our estimation methodologies under exogenous regressors and endogenous regressors, respectively. Section 7 presents the Monte Carlo simulation. Section 8 gives the empirical results. All mathematical proofs are collected in Appendix.

2. High frequency factor model with partially observable factors

We consider the following high frequency factor model:

$$d\mathbf{Y}_t = \mathbf{B}_t^\top d\mathbf{F}_t^\circ + \sum_{k=1}^{d+q} \int_{\mathbb{R}} \mathbf{B}_t^{J, (k)} x \tilde{\mu}_{\mathbf{F}}^{(k)}(dt, dx) + d\mathbf{U}_t, \quad (2.1)$$

where $\mathbf{Y}_t = (Y_t^{(1)}, Y_t^{(2)}, \dots, Y_t^{(p)})^\top$ is the dependent variable process, $\mathbf{F}_t = (F_t^{(1)}, F_t^{(2)}, \dots, F_t^{(d+q)})^\top$ is the factor process, and “ $\circ$ ” denotes the continuous part of the corresponding stochastic process, $\mathbf{B}_t = \left\{ \mathbf{B}_t^{(r, s)} \right\}_{1 \leq r \leq d+q, 1 \leq s \leq p}$ is the continuous factor loading process, $\mathbf{B}_t^J = (\mathbf{B}_t^{J, (1)}, \mathbf{B}_t^{J, (2)}, \dots, \mathbf{B}_t^{J, (d+q)})^\top$ with $\mathbf{B}_t^{J, (i)} = (B_t^{J, (i, 1)}, B_t^{J, (i, 2)}, \dots, B_t^{J, (i, p)})^\top$ is the discontinuous factor loading process, $\tilde{\mu}_{\mathbf{F}}^{(k)}$ denotes the jump measure corresponding to the k th factor $F_t^{(k)}$, and $\mathbf{U}_t = (U_t^{(1)}, U_t^{(2)}, \dots, U_t^{(p)})^\top$ is the idiosyncratic component process. Here the positive integer d denotes the number of observable factors while q denotes the number of latent factors. The setting in Eq. (2.1) is quite flexible in two-fold: (i) it can distinguish the continuous and discontinuous factor loading (or, beta), which is evident in literature, see, e.g., Todorov and Bollerslev (2010), Bollerslev et al. (2016); (ii) it allows the factor loading process to have time-varying feature as documented in Andersen et al. (2021, 2023). In this paper, we mainly consider the time-varying evolution of the continuous part in the factor structure.

Because the factors in Eq. (2.1) are partially observable, we can further write $\mathbf{B}_t$, $\mathbf{B}_t^J$ and $\mathbf{F}_t$ as follows:

$$\mathbf{B}_t^\top = (\beta_t^\top, \theta_t^\top), (\mathbf{B}_t^J)^\top = ((\beta_t^J)^\top, (\theta_t^J)^\top), \text{ and } \mathbf{F}_t^\top = (\mathbf{X}_t^\top, \mathbf{L}_t^\top), \quad (2.2)$$

where $\mathbf{X}_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(d)})^\top$ is the *observable* factor process; $\beta_t = \left\{ \beta_t^{(r, s)} \right\}_{1 \leq r \leq d, 1 \leq s \leq p}$ and $\beta_t^J = \left\{ \beta_t^{J, (r, s)} \right\}_{1 \leq r \leq d, 1 \leq s \leq p}$ are the continuous and discontinuous factor loading processes for the observable factors, respectively; $\mathbf{L}_t = (L_t^{(1)}, L_t^{(2)}, \dots, L_t^{(q)})^\top$ is the *latent* factor process; $\theta_t = \left\{ \theta_t^{(r, s)} \right\}_{1 \leq r \leq q, 1 \leq s \leq p}$ and $\theta_t^J = \left\{ \theta_t^{J, (r, s)} \right\}_{1 \leq r \leq q, 1 \leq s \leq p}$ are the continuous and discontinuous factor loading processes for the latent factors, respectively.

Based on the partition in Eq. (2.2), we could further write Eq. (2.1) as follows:

$$d\mathbf{Y}_t = \beta_t^\top d\mathbf{X}_t^\circ + \theta_t^\top d\mathbf{L}_t^\circ + \sum_{k=1}^{d+q} \int_{\mathbb{R}} \mathbf{B}_t^{J, (k)} x \tilde{\mu}_{\mathbf{F}}^{(k)}(dt, dx) + d\mathbf{U}_t, \quad (2.3)$$

and when we only consider the continuous parts of the above formula, it can be written as follows:

$$d\mathbf{Y}_t^\circ = \beta_t^\top d\mathbf{X}_t^\circ + \theta_t^\top d\mathbf{L}_t^\circ + d\mathbf{U}_t^\circ. \quad (2.4)$$

To estimate this partially observable factor model, we usually first conduct the high frequency regression for the observable factors ($\mathbf{X}_t$). For the ease of illustration of the high frequency regression, we define the corresponding residual process in high frequency regression as follows:

$$\mathbf{Z}_t^\circ = (Z_t^{(1)}, Z_t^{(2)}, \dots, Z_t^{(p)})^\top,$$

and

$$d\mathbf{Z}_t^\circ = \theta_t^\top d\mathbf{L}_t^\circ + d\mathbf{U}_t^\circ, \quad (2.5)$$

and consequently, Eq. (2.4) can be further written as:

$$d\mathbf{Y}_t^\circ = \beta_t^\top d\mathbf{X}_t^\circ + d\mathbf{Z}_t^\circ. \quad (2.6)$$

In general, we assume $\mathbf{X}_t$, $\mathbf{L}_t$, and $\mathbf{U}_t$ are Itô semimartingales satisfying standard regularity assumptions, which will be rigorously stated in subsequent section. Here we denote the continuous parts of $\mathbf{Y}_t$, $\mathbf{X}_t$, $\mathbf{L}_t$, $\mathbf{U}_t$ and $\mathbf{Z}_t$ by $\mathbf{Y}_t^\circ$, $\mathbf{X}_t^\circ$, $\mathbf{L}_t^\circ$, $\mathbf{U}_t^\circ$ and $\mathbf{Z}_t^\circ$, respectively.

Next, we assume the orthogonality between the common factors $\mathbf{F}_t$ and idiosyncratic components $\mathbf{U}_t$.

Assumption 1. [Assumption on the Orthogonality between $\mathbf{F}$ and $\mathbf{U}$] For arbitrary $0 \leq t \leq \mathcal{T}$, we assume the following orthogonality condition:

$$[\mathbf{X}^\circ, \mathbf{U}^\circ]_t = 0, \quad (2.7)$$

$$[\mathbf{L}^\circ, \mathbf{U}^\circ]_t = 0. \quad (2.8)$$

3. Basic setup

3.1. Data generating process of efficient prices

Recall the model of high frequency regression as defined in Eq. (2.6). In what follows, we provide the detailed assumptions on factors, continuous factor loadings and idiosyncratic components.

Assumption 2 (Assumptions on Factors, Continuous Factor Loadings and Idiosyncratic Components).

- (i). The observable factor process $\mathbf{X}_t$, the latent factor process $\mathbf{L}_t$, and the residual process $\mathbf{U}_t$ are d -dimensional, q -dimensional, and p -dimensional Itô semimartingales in the sense of Definition A.1 in Appendix A, respectively.
- (ii). The continuous factor loading processes (β_t) and (θ_t) are predictable and locally bounded.

Remark 1. *Assumption 2* assumes that the cross-sectional high frequency price process $(\mathbf{Y}_t)$ has finite variation jumps, i.e., $\varsigma \in [0, 1)$ as stated in Definition A.1. However, it is necessary to mention that the infinite variation jumps in the price process are well documented in the literature of high frequency econometrics, i.e., [Aït-Sahalia and Jacod \(2009\)](#), [Jing et al. \(2012\)](#). Testing procedures have been further developed for infinite variation jumps, see, e.g., [Kong et al. \(2015\)](#), [Kong \(2019\)](#). Under the assumption of infinite variation jumps, the estimation theory on integrated volatility functional has been developed by [Jacod and Todorov \(2014, 2016\)](#) for the non-noisy observations. As a necessary building block of noise-robust estimation methods of integrated volatility functional, [Liu and Liu \(2024\)](#) developed spot volatility estimation under both infinite variation jumps and dependent microstructure noise. The formal establishment of the noise-robust estimation methods for integrated volatility functional with infinite variation jumps will be left for future research.

Here we provide notations for several spot covariance matrices.

Definition 1 (Definitions of Spot Covariances). For all $0 \leq t \leq \mathcal{T}$, we define:

- (i) Spot covariance of observable factors: $c_t^{\mathbf{X}, \mathbf{X}} \triangleq \frac{d}{dt} [\mathbf{X}^\circ, \mathbf{X}^\circ]_t$,
- (ii) Inverse spot covariance of observable factors: $\mathbf{A}_t = (c_t^{\mathbf{X}, \mathbf{X}})^{-1}$,
- (iii) Spot covariance of latent factors: $c_t^{\mathbf{L}, \mathbf{L}} \triangleq \frac{d}{dt} [\mathbf{L}^\circ, \mathbf{L}^\circ]_t$,
- (iv) Spot idiosyncratic covariance matrix: $\mathbf{r}_t \triangleq \frac{d}{dt} [\mathbf{U}^\circ, \mathbf{U}^\circ]_t$,
- (v) Spot residual covariance matrix: $\mathbf{s}_t \triangleq \frac{d}{dt} [\mathbf{Z}^\circ, \mathbf{Z}^\circ]_t$,
- (vi) Spot covariance between observable factors and latent factors: $c_t^{\mathbf{X}, \mathbf{Y}} \triangleq \frac{d}{dt} [\mathbf{X}^\circ, \mathbf{Y}^\circ]_t \triangleq (c_t^{\mathbf{Y}, \mathbf{X}})^\top$,
- (vii) Spot covariance of dependent variable processes: $c_t^{\mathbf{Y}, \mathbf{Y}} \triangleq \frac{d}{dt} [\mathbf{Y}^\circ, \mathbf{Y}^\circ]_t$,

where $c_t^{\mathbf{Y}, \mathbf{Y}}$ is a $p \times p$ matrix process, $c_t^{\mathbf{X}, \mathbf{Y}}$ is a $d \times p$ matrix process and $c_t^{\mathbf{Y}, \mathbf{X}}$ is a $p \times d$ vector process.

Because only $(\mathbf{Y}_t)$ and $(\mathbf{X}_t)$ are observable in this partially observable factor model, we construct a vector process which including both $(\mathbf{Y}_t)$ and $(\mathbf{X}_t)$ as

$$\Theta_t = (\mathbf{Y}_t^\top, \mathbf{X}_t^\top)^\top \text{ with } \mathbf{Y}_t = (Y_t^{(1)}, Y_t^{(2)}, \dots, Y_t^{(p)})^\top \text{ and } \mathbf{X}_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(d)})^\top,$$

where it is straightforward to see that Θ_t is a $(p + d)$ -Itô semimartingale in the sense of Definition A.1 in Appendix A. Denote the continuous part of Θ_t by Θ_t° . The spot covariance of Θ_t° can be decomposed as:

$$c_t = [\Theta^\circ, \Theta^\circ]_t' = \begin{bmatrix} c_t^{\mathbf{Y}, \mathbf{Y}} & c_t^{\mathbf{Y}, \mathbf{X}} \\ c_t^{\mathbf{X}, \mathbf{Y}} & c_t^{\mathbf{X}, \mathbf{X}} \end{bmatrix}.$$

Assumption 3 (Assumptions on the Spot Covariance Matrix c_t). The spot covariance matrix process c_t satisfies that $\sup_{0 \leq t \leq \mathcal{T}} \|c_t\|_{\max} < \infty$ and $\inf_{0 \leq t \leq \mathcal{T}} \lambda_{\min}(c_t) > 0$.

3.2. Data description

In this paper, we consider the noisy and asynchronous/irregular high frequency observations. The detailed description of the observed data is as follows.

Each element of the log stock price process, i.e., $\Theta_t^{(r)}$, $1 \leq r \leq p + d$, is observed on the asynchronous grid:

$$\mathcal{G}^{(r)} = \left\{ 0 = t_0^{(r)} < t_1^{(r)} < \dots < t_{n^{(r)}}^{(r)} = \mathcal{T} \right\}.$$

The observed process $\Theta^* = (\Theta^{*,(1)}, \Theta^{*,(2)}, \dots, \Theta^{*,(p+d)})$ is contaminated by the market microstructure noise ϵ as follows:

$$\Theta_{t_j^{(r)}}^{*(r)} = \Theta_{t_j^{(r)}}^{(r)} + \epsilon_{t_j^{(r)}}^{(r)}, \text{ for } r = 1, 2, \dots, p + d \text{ and } 0 \leq j \leq n^{(r)}. \quad (3.1)$$

The general assumption about the market microstructure noise and the observation times are as follows.

Assumption 4 (*Assumption on Noise and Observation Time*). We assume the observation times $t_j^{(r)}$ are $(\mathcal{F}_t)$ -stopping times for all $1 \leq r \leq p + d$. For each r and j , the noise $\epsilon_{t_j^{(r)}}^{(r)}$ is observed at time $t_j^{(r)}$ (i.e., is $\mathcal{F}_{t_j^{(r)}}$ -measurable), and $\sup_{r,j} E \left(\epsilon_{t_j^{(r)}}^{(r)} \right)^2 < \infty$ and $E \epsilon_{t_j^{(r)}}^{(r)} = 0$.

4. Estimator of spot covariance

We define the estimator of spot covariance $c_t^{(r,s)}$ as follows: for $1 \leq r, s \leq p + d$,

$$\hat{c}_{\Delta T_n, t}^{(r,s)} = \frac{1}{\Delta T_n} \left(\langle \widehat{\Theta^{(r)}}, \widehat{\Theta^{(s)}} \rangle_{t+\Delta T_n} - \langle \widehat{\Theta^{(r)}}, \widehat{\Theta^{(s)}} \rangle_t \right), \quad (4.1)$$

where $\langle \widehat{\Theta^{(r)}}, \widehat{\Theta^{(s)}} \rangle_t$ is the Truncated S-TSRV estimator for the integrated covariance $\int_0^t c_u^{(r,s)} du$ with the convergence rate a_n , i.e.,

$$\langle \widehat{\Theta^{(r)}}, \widehat{\Theta^{(s)}} \rangle_t = \int_0^t c_u^{(r,s)} du + M_{n,t}^{(r,s)} + o_p(a_n) \text{ for } \forall t \in [0, \mathcal{T}]. \quad (4.2)$$

The detailed definition and assumptions on the Truncated S-TSRV are provided in Appendix B for the conciseness of discussion. For example, the explicit formulas for $M_{n,t}^{(r,s)}$ and a_n can be found in Appendix B, i.e., Eqn. (B.5) and Eq. (4.3), respectively. Note that $M_{n,t}^{(r,s)}$ is a martingale with statistical order of a_n . In general, the convergence rate a_n of the Truncated S-TSRV has the expression of

$$a_n = [(K_n - J_n) \Delta \tau_n]^{\frac{1}{2}}. \quad (4.3)$$

where $K_n - J_n$ is the difference of two time scales and $\Delta \tau_n$ denotes the length of pre-averaging window used in Truncated S-TSRV. Note that a_n can be arbitrarily close to $n^{-1/4}$ based on the discussion in Remark 5 of Chen (2024).

In addition, we assume $\{\Delta T_n\}_{n \geq 1}$ is a sequence of positive numbers satisfying

$$a_n^{-2} \Delta T_n \rightarrow \infty \text{ and } \Delta T_n \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (4.4)$$

Based on the partition of Θ_t , i.e., $\Theta_t = (\mathbf{Y}_t^\top, \mathbf{X}_t^\top)^\top$, the spot covariance estimate $\hat{c}_{\Delta T_n, t} = \left\{ \hat{c}_{\Delta T_n, t}^{(r,s)} \right\}_{1 \leq r, s \leq p+d}$ can be further partitioned into the following sample analogs:

$$\hat{c}_{\Delta T_n, t} = \begin{bmatrix} \hat{c}_{\Delta T_n, t}^{\mathbf{Y}, \mathbf{Y}} & \hat{c}_{\Delta T_n, t}^{\mathbf{Y}, \mathbf{X}} \\ \hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{Y}} & \hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{X}} \end{bmatrix} \quad (4.5)$$

where $\hat{c}_{\Delta T_n, t}^{\mathbf{Y}, \mathbf{Y}}$ is a $p \times p$ matrix, $\hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{X}}$ is a $d \times d$ matrix, $\hat{c}_{\Delta T_n, t}^{\mathbf{Y}, \mathbf{X}}$ is a $p \times d$ matrix, which satisfying $\hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{X}} = \left(\hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{X}} \right)^\top$ and $\hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{Y}} = \left(\hat{c}_{\Delta T_n, t}^{\mathbf{Y}, \mathbf{X}} \right)^\top$. We further define

$$\hat{\mathbf{A}}_{\Delta T_n, t} = \left(\hat{c}_{\Delta T_n, t}^{\mathbf{X}, \mathbf{X}} \right)^{-1}. \quad (4.6)$$

4.1. Some necessary notations

To facilitate the construction of related estimators, we create a new equidistant grid:

$$T_{n,i} = i \Delta T_n, \text{ for } 1 \leq i \leq B_n, \quad (4.7)$$

where $B_n = \mathcal{T} / \Delta T_n$ and ΔT_n follows the assumption in Eq. (4.4).

To develop the bias correction term and the estimator of asymptotic variance, for $1 \leq r, s \leq p + d$, we define

$$\check{\phi}_{\Delta T_n, T_{n,i-1}}^{(r,s)} = \frac{1}{2} \left(\hat{c}_{\Delta T_n/2, (i-1/2)\Delta T_n}^{(r,s)} - \hat{c}_{\Delta T_n/2, (i-1)\Delta T_n}^{(r,s)} \right).$$

Then we have the following representations:

$$\check{\phi}_{\Delta T_n, T_{n,i-1}} = \begin{bmatrix} \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{Y}} & \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{X}} \\ \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{Y}} & \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \end{bmatrix}_{(p+d) \times (p+d)} \text{ and } \check{\phi}_{\Delta T_n, T_{n,i-1}}^* = \begin{bmatrix} \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{X}} \\ \check{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \end{bmatrix}_{(p+d) \times d}. \quad (4.8)$$

5. Uncorrelated observable & latent factors: High frequency factor analysis with partially observable factors under finite dimensionality

For the purpose of identification, we assume the following orthogonality condition.

Assumption 5. (Assumption on the Orthogonality between $\mathbf{X}$ and $\mathbf{L}$) For arbitrary $0 \leq t \leq \mathcal{T}$, we assume the following orthogonality condition:

$$[\mathbf{X}^\circ, \mathbf{L}^\circ]_t = 0. \quad (5.1)$$

Based on Eq. (2.5) and Assumption 1, we know that the observed factors are exogenous in the high frequency regression, i.e., Eq. (2.6), i.e.,

$$[\mathbf{X}^\circ, \mathbf{Z}^\circ]_t = 0. \quad (5.2)$$

Under Assumptions 1 and 5, we obtain the following identify:

$$c_t^{\mathbf{Y},\mathbf{X}} = \beta_t^\top c_t^{\mathbf{X},\mathbf{X}}, \quad (5.3)$$

which yields the expression of the ordinary least squares (OLS) solution for the observable factor loading as follows:

$$\beta_t^{\text{OLS}} = \left(c_t^{\mathbf{X},\mathbf{X}}\right)^{-1} c_t^{\mathbf{X},\mathbf{Y}}, \quad (5.4)$$

provided $c_t^{\mathbf{X},\mathbf{X}}$ is invertible.

Based on Assumptions 1 and 5, we have:

$$c_t^{\mathbf{Y},\mathbf{Y}} = (\beta_t^{\text{OLS}})^\top c_t^{\mathbf{X},\mathbf{X}} \beta_t^{\text{OLS}} + \mathbf{s}_t, \quad (5.5)$$

$$\mathbf{s}_t = \theta_t^\top c_t^{\mathbf{L},\mathbf{L}} \theta_t + \mathbf{r}_t, \quad (5.6)$$

which implies that

$$c_t^{\mathbf{Y},\mathbf{Y}} = (\beta_t^{\text{OLS}})^\top c_t^{\mathbf{X},\mathbf{X}} \beta_t^{\text{OLS}} + \theta_t^\top c_t^{\mathbf{L},\mathbf{L}} \theta_t + \mathbf{r}_t. \quad (5.7)$$

Plugging Eq. (5.4) into above identities, we have:

$$c_t^{\mathbf{Y},\mathbf{Y}} = c_t^{\mathbf{Y},\mathbf{X}} \left(c_t^{\mathbf{X},\mathbf{X}}\right)^{-1} c_t^{\mathbf{X},\mathbf{Y}} + \mathbf{s}_t.$$

Then the spot covariance of θ_t° can be expressed as:

$$c_t = \begin{bmatrix} (\beta_t^{\text{OLS}})^\top c_t^{\mathbf{X},\mathbf{X}} \beta_t^{\text{OLS}} + \mathbf{s}_t & (\beta_t^{\text{OLS}})^\top c_t^{\mathbf{X},\mathbf{X}} \\ c_t^{\mathbf{X},\mathbf{X}} \beta_t^{\text{OLS}} & c_t^{\mathbf{X},\mathbf{X}} \end{bmatrix} = \begin{bmatrix} c_t^{\mathbf{Y},\mathbf{X}} \left(c_t^{\mathbf{X},\mathbf{X}}\right)^{-1} c_t^{\mathbf{X},\mathbf{Y}} + \mathbf{s}_t & c_t^{\mathbf{Y},\mathbf{X}} \\ c_t^{\mathbf{X},\mathbf{Y}} & c_t^{\mathbf{X},\mathbf{X}} \end{bmatrix},$$

where $c_t^{\mathbf{Y},\mathbf{Y}}$, $c_t^{\mathbf{X},\mathbf{X}}$, $c_t^{\mathbf{Y},\mathbf{X}}$, and $c_t^{\mathbf{X},\mathbf{Y}}$ are defined in Definition 1, which satisfying $c_t^{\mathbf{X},\mathbf{X}} = \left(c_t^{\mathbf{X},\mathbf{X}}\right)^\top$ and $c_t^{\mathbf{Y},\mathbf{X}} = \left(c_t^{\mathbf{X},\mathbf{Y}}\right)^\top$.

Under finite dimensionality, i.e., d and p are finite in asymptotics, we first conduct the high frequency regression to handle the observable factors and secondly we conduct the high frequency PCA to detect the latent factors. In the first step, we mainly state the results about the estimation of the residual covariance while in the second step, we propose the estimation methodology for the integrated eigenvalues of the residual covariance matrix.

5.1. Step 1. High frequency regression on the observable factors

As the first step, we conduct high frequency regression on the observable factors, and obtain the representation of the residual as follows:

$$\mathbf{d}\hat{\mathbf{Z}}_t^{\circ,\text{OLS}} = \mathbf{d}\mathbf{Y}_t^\circ - \beta_t^{\text{OLS}} \mathbf{d}\mathbf{X}_t^\circ = (\beta_t - \beta_t^{\text{OLS}})^\top \mathbf{d}\mathbf{X}_t^\circ + \theta_t^\top \mathbf{d}\mathbf{L}_t^\circ + \mathbf{d}\mathbf{U}_t^\circ, \quad (5.8)$$

The exogeneity of observation factors, i.e., Assumption 5, implies that the identity $\beta_t = \beta_t^{\text{OLS}}$ for all $0 \leq t \leq \mathcal{T}$, and thus, the residual term follows a purely latent factor structure:

$$\mathbf{d}\hat{\mathbf{Z}}_t^{\circ,\text{OLS}} = \theta_t^\top \mathbf{d}\mathbf{L}_t^\circ + \mathbf{d}\mathbf{U}_t^\circ, \quad (5.9)$$

which motivates us to apply the high frequency PCA on the residual process $\mathbf{d}\hat{\mathbf{Z}}_t^{\circ,\text{OLS}}$.

In the scenario of high frequency regression, based on the orthogonality $[\mathbf{X}^\circ, \mathbf{Z}^\circ]_t = 0$, the estimator for the spot residual covariance can be expressed as follows:

$$\hat{\mathbf{s}}_{\Delta T_{n,t}} = \hat{c}_{\Delta T_{n,t}}^{\mathbf{Y},\mathbf{Y}} - (\hat{\beta}_{\Delta T_{n,t}}^{\text{OLS}})^\top \hat{c}_{\Delta T_{n,t}}^{\mathbf{X},\mathbf{X}} \hat{\beta}_{\Delta T_{n,t}}^{\text{OLS}}, \quad (5.10)$$

where the spot OLS beta estimate $\hat{\beta}_{\Delta T_n, t}^{\text{OLS}}$ is defined as follows, i.e.,

$$\hat{\beta}_{\Delta T_n, t}^{\text{OLS}} = \left(\hat{c}_{\Delta T_n, t}^{\text{X,X}} \right)^{-1} \hat{c}_{\Delta T_n, t}^{\text{X,Y}}, \quad (5.11)$$

and $\hat{c}_{\Delta T_n, t}^{\text{X,Y}}$ and $\hat{c}_{\Delta T_n, t}^{\text{X,X}}$ are defined in Eq. (4.5). In the following lemma, we state the elementwise consistency of $\hat{s}_{\Delta T_n, t}$.

Lemma 1. Suppose Assumptions 1–5 and further assume $a_n^{-2} \Delta T_n \rightarrow \infty$ and $\Delta T_n \rightarrow 0$ as $n \rightarrow \infty$. For the setting of Truncated S-TSRV, suppose Assumptions B.1–B.3 as stated in Appendix B. Suppose d and p are finite in asymptotics. Then we obtain that for $1 \leq l, m \leq p$ and arbitrary ϵ satisfying $0 < \epsilon < 1/2$,

$$\sup_i \left\| \hat{s}_{\Delta T_n, T_{n,i-1}}^{(l,m)} - s_{T_{n,i-1}}^{(l,m)} \right\|_2 = O_p \left(\Delta T_n^{1/2-\epsilon} \right) + O_p \left((a_n^2 \Delta T_n^{-1})^{1/2-\epsilon} \right) = o_p(1).$$

This proof of this lemma is similar to that of Lemma 2 in Chen et al. (2023).

5.2. Step II. High frequency PCA on the residual covariance

Since the residual process follows a purely latent factor structure, i.e., Eq. (5.8), the high frequency PCA on residual process will result in the Residual Principal Components (hereafter, Residual PCs), which can be expressed as follows:

$$\int_0^T [\gamma^{(l)}(s_t)]^\top d\mathbf{Z}_t^\circ, \quad (5.12)$$

where $\gamma^{(l)}(\mathbf{A})$ denotes the eigenvector of some positive semi-definite matrix $\mathbf{A}$ corresponding to its l th largest eigenvalue $\lambda^{(l)}(\mathbf{A})$. It is straightforward to see that the quadratic variation of the l th residual PC is the integrated eigenvalues of the residual covariance matrix, i.e.,

$$\int_0^T \lambda^{(l)}(s_t) dt, \quad (5.13)$$

of which the estimation problem can be reduced to the estimation of integrated spectral functions.

In this section, we mainly develop the estimation methodology of the integrated eigenvalues of the residual covariance matrix s_t . Under the scenario of finite dimensionality, the estimation of integrated eigenvalues can be deduced into the estimation of integrated spectral functions.

For $0 \leq t \leq T$, suppose the $p \times p$ residual covariance matrix s_t to be positive semi-definite and assume that the eigenvalues of s_t satisfy:

$$\lambda^{(1)}(s_t) = \dots = \lambda^{(g_1)}(s_t) > \lambda^{(g_1+1)}(s_t) = \dots = \lambda^{(g_2)}(s_t) > \dots > \lambda^{(g_{r-1})}(s_t) > \lambda^{(g_{r-1}+1)}(s_t) = \lambda^{(g_r)}(s_t) \geq 0,$$

where $g_r = p$, and r is the number of distinct eigenvalues. We would like to estimate the following integrated spectral functions:

$$\int_0^T F^\lambda(s_t) dt,$$

where

$$F^\lambda(\cdot) = \left(\frac{1}{g_1} \sum_{j=1}^{g_1} \lambda^{(j)}(\cdot), \frac{1}{g_2 - g_1} \sum_{j=g_1+1}^{g_2} \lambda^{(j)}(\cdot), \dots, \frac{1}{g_r - g_{r-1}} \sum_{j=g_{r-1}+1}^{g_r} \lambda^{(j)}(\cdot) \right)^\top. \quad (5.14)$$

We could also write $F^\lambda(\cdot)$ using its components: $F_l^\lambda(\cdot)$ with $l = 1, 2, \dots, r$.

Note that a spectral function F defined on the $p \times p$ positive semi-definite matrix $\mathcal{X}$ can be expressed as the composition of some symmetric function f with an open symmetric domain on $\mathbb{R}_p^+$ and the non-increasing sequence of the eigenvalues of $\mathcal{X}$, i.e., $F = f \circ \lambda$ where $\circ$ denotes the function composition. Also note that $f \in C^l$ at $\lambda(\mathcal{X})$ is equivalent to $F \in C^l$ at $\mathcal{X}$, for any non-negative integer l .

The symmetric function $f_l(x)$ corresponding to this spectral function $F_l^\lambda(X)$ is:

$$f_l(x) = \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} x^{(j)},$$

where $x^{(k)}$ denotes k th largest element of the vector $x = (x^{(1)}, x^{(2)}, \dots, x^{(p)})$. Indeed, $f_l(x) \in C^\infty$ at any point $x = (x^{(1)}, x^{(2)}, \dots, x^{(p)}) \in \mathbb{R}_p^+$ with $x^{(g_{l-1})} > x^{(g_{l-1}+1)} = \dots = x^{(g_l)} > x^{(g_l+1)}$ and $\partial f_l(x) = \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} e_j$ and $\partial^h f_l(x) = 0$ for $h \geq 2$. Thus, it is easy to see that f_l is C^3 on $\mathbf{D}(g_1, g_2, \dots, g_r)$, where $\mathbf{D}(g_1, g_2, \dots, g_r)$ denotes a symmetric open subset of $\mathbb{R}_p^+ / \{\mathbf{0}\}$ which satisfying

$$\mathbf{D}(g_1, g_2, \dots, g_r) = \left\{ x \in \mathbb{R}_p^+ / \{\mathbf{0}\} \mid x^{(g_l)} > x^{(g_l+1)} \text{ for any } l = 1, 2, \dots, r-1 \right\}.$$

Based on the theory of spectral function, see, i.e., [Aït-Sahalia and Xiu \(2019\)](#), we have the following derivatives of the function F_l^λ with respect to some p -dimensional positive semi-definite matrix $\mathcal{X}$:

$$\partial_{rs} F_l^\lambda(\mathcal{X}) = \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} O^{(j,r)} O^{(j,s)}, \quad (5.15a)$$

$$\partial_{r_1 s_1, r_2 s_2}^2 F_l^\lambda(\mathcal{X}) = \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} \left(O^{(j,s_1)} P_j^{(r_1, s_2)} O^{(j,s_2)} + O^{(j,r_1)} P_j^{(s_1, s_2)} O^{(j,r_2)} \right), \quad (5.15b)$$

where O is any orthogonal matrix such that $\mathcal{X} = O^\top \text{Diag}(\lambda(\mathcal{X})) O$,

$$P_j \triangleq (\lambda^{(j)}(\mathcal{X}) \mathbb{I}_p - \mathcal{X})^+, \quad (5.16)$$

$\mathbb{I}_p$ is the p -dimensional identity matrix and the superscript “+” denotes the Moore–Penrose inverse of a real matrix.

For any $1 \leq g_1 < g_2 < \dots < g_r \leq p$, define the set

$$\mathbf{S}(g_1, g_2, \dots, g_r) = \left\{ \mathcal{X} \in S_d^{++} \mid \lambda_{g_l}(\mathcal{X}) > \lambda_{g_l+1}(\mathcal{X}) \text{ for any } l = 1, 2, \dots, r-1 \right\}.$$

Define a subset of $\mathbf{S}(g_1, g_2, \dots, g_r)$, where the residual covariance matrix $\mathbf{s}_l$ will take values, i.e.,

$$\begin{aligned} \mathbf{S}^*(g_1, g_2, \dots, g_r) &= \{ \mathcal{X} \in S_d^{++} \mid \lambda_1(\mathcal{X}) = \dots = \lambda_{g_1}(\mathcal{X}) > \lambda_{g_1+1}(\mathcal{X}) = \dots \\ &= \lambda_{g_2}(\mathcal{X}) > \dots > \lambda_{g_{r-1}}(\mathcal{X}) > \lambda_{g_{r-1}+1}(\mathcal{X}) = \dots = \lambda_{g_r}(\mathcal{X}) \}. \end{aligned}$$

The assumptions on the residual covariance matrix $\mathbf{s}_l$ can be stated as follows.

Assumption 6. There exists some open and convex set C , such that its closure $\bar{C} \subset \mathbf{S}(g_1, g_2, \dots, g_r)$, where $1 \leq g_1 < g_2 < \dots < g_r \leq d$, and that for any $0 \leq t \leq \mathcal{T}$, $\mathbf{s}_t \in C \cap \mathbf{S}^*(g_1, g_2, \dots, g_r)$.

By letting $g_0 = 0$, for $l = 1, 2, \dots, r$, denote the l th component of $F^\lambda(\mathbf{s}_t)$ by $\mathfrak{F}_t^{(l)}$, i.e.,

$$\mathfrak{F}_t^{(l)} \triangleq F_l^\lambda(\mathbf{s}_t) = \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} \lambda^{(j)}(\mathbf{s}_t). \quad (5.17)$$

In what follows, we state the main results of the estimation methodology for

$$\mathfrak{V}_T^{(l)} \triangleq \int_0^T \mathfrak{F}_t^{(l)} dt, \quad l = 1, 2, \dots, r. \quad (5.18)$$

5.3. Theoretical results for $\int_0^T \mathfrak{F}_t^{(l)} dt$ estimation

The bias-corrected estimator is defined as:

$$\tilde{\mathfrak{V}}_T^{(l)} \triangleq \Delta T_n \sum_{i=1}^{B_n} \left[\hat{\mathfrak{F}}_{T_{n,i-1}}^{(l)} - \hat{\mathfrak{B}}_{T_{n,i-1}}^{(l)} \right], \quad (5.19)$$

where

$$\hat{\mathfrak{F}}_{T_{n,i-1}}^{(l)} \triangleq F_l^\lambda(\hat{\mathbf{s}}_{\Delta T_n, T_{n,i-1}}), \quad (5.20)$$

$$\hat{\mathfrak{B}}_{T_{n,i-1}}^{(l)} \triangleq \frac{1}{g_l - g_{l-1}} \sum_{j=g_{l-1}+1}^{g_l} [\hat{O}_{\Delta T_n, T_{n,i-1}} \Psi_{\Delta T_n, T_{n,i-1}}^j \hat{O}_{\Delta T_n, T_{n,i-1}}^\top]^{(j,j)}, \quad (5.21)$$

and the spot residual covariance estimator $\hat{\mathbf{s}}_{\Delta T_n, T_{n,i-1}}$ is defined in Eq. (5.10), and $\hat{O}_{\Delta T_n, T_{n,i-1}}$ satisfies that

$$\hat{\mathbf{s}}_{\Delta T_n, T_{n,i-1}} = \hat{O}_{\Delta T_n, T_{n,i-1}}^\top \text{Diag}(\lambda(\hat{\mathbf{s}}_{\Delta T_n, T_{n,i-1}})) \hat{O}_{\Delta T_n, T_{n,i-1}}, \quad (5.22)$$

and $\Psi_{\Delta T_n, T_{n,i-1}}^j$ is defined in Eqn. (E.3) of Appendix E.1.2. The estimator for the asymptotic variance is defined as:

$$\widehat{\text{AVAR}}(\tilde{\mathfrak{V}}_T^{(l)}) \triangleq \frac{\Delta T_n^2}{(g_l - g_{l-1})^2} \sum_{i=1}^{B_n} \left(\sum_{j=g_{l-1}+1}^{g_l} [\hat{O}_{\Delta T_n, T_{n,i-1}} \hat{\boldsymbol{\eta}}_{\Delta T_n, T_{n,i-1}} \hat{O}_{\Delta T_n, T_{n,i-1}}^\top]^{(j,j)} \right)^2, \quad (5.23)$$

where $\hat{O}_{\Delta T_n, T_{n,i-1}}$ is defined in Eq. (5.22), and $\hat{\boldsymbol{\eta}}_{\Delta T_n, T_{n,i-1}}$ is defined in Eqn. (E.3) of Appendix E.1.2.

Finally, we state the central limit theorem for $\tilde{\mathfrak{V}}_T^{(l)}$ as follows.

Theorem 1. Suppose [Assumptions 1–6](#) and further assume $a_n^{-1}\Delta T_n \rightarrow 0$ and $a_n^{-3/2}\Delta T_n \rightarrow \infty$ as $n \rightarrow \infty$. For the setting of Truncated S-TSRV, suppose [Assumptions B.1–B.3](#) as stated in [Appendix B](#). Suppose d and p are finite in asymptotics. Then for $1 \leq l \leq r$, we obtain:

$$\frac{\tilde{\mathfrak{Y}}_{\mathcal{T}}^{(l)} - \mathfrak{Y}_{\mathcal{T}}^{(l)}}{\sqrt{\widehat{AVAR}(\tilde{\mathfrak{Y}}_{\mathcal{T}}^{(l)})}} \xrightarrow{\mathcal{L}} N(0, 1)$$

stably.

The proof of theorem is provided in [Appendix G](#).

6. Correlated observable & latent factors: High frequency unsupervised exogenous component learning

In this section, we assume that the observable factors and the latent factors can be correlated. Therefore, we will derive the statistical inference without imposing [Assumption 5](#). In this case, the observable factors $\mathbf{X}$ in the high frequency regression, see, i.e., [Eq. \(2.6\)](#), becomes endogenous regressors. In the literature of statistics and econometrics, instrumental variables could be introduced in order to deal with the problem of endogenous regressors. Motivated by the idea of classical instrumental variable approach, we propose the high frequency unsupervised exogenous component learning (HF-UECL) to carry out the factor analysis when the observable factor and latent factors are correlated. Note that the term “unsupervised” means that the exogenous components are not directly observable.

6.1. High frequency exogenous component

In this subsection, we state the related results about high frequency exogenous component.

Suppose $\mathbf{V}_t = \left(V_t^{(1)}, V_t^{(2)}, \dots, V_t^{(\tilde{d})} \right)^\top$ is the exogenous component process corresponding to the observable factors $\mathbf{X}$, i.e.,

$$d\mathbf{X}_t = \delta_t^\top d\mathbf{V}_t + d\mathbf{D}_t, \quad (6.1)$$

where $\tilde{d}$ is known and $\tilde{d} \leq d$, $\mathbf{D}_t = \left(D_t^{(1)}, D_t^{(2)}, \dots, D_t^{(d)} \right)^\top$ is the error term, $\delta_t = \left\{ \delta_t^{(r,s)} \right\}_{1 \leq r \leq \tilde{d}, 1 \leq s \leq d}$ is the loadings of $\mathbf{V}_t$ with respect to $\mathbf{X}$. In general, we assume $\mathbf{V}_t$, $\mathbf{D}_t$, and δ_t are Itô semimartingales satisfying standard regularity assumptions. We denote the continuous parts of $\mathbf{V}_t$ and $\mathbf{D}_t$ by $\mathbf{V}_t^\circ$ and $\mathbf{D}_t^\circ$, respectively.

In the following assumption, we impose related orthogonality conditions for exogenous component $\mathbf{V}_t$.

Assumption 7. [Assumption on the orthogonality related to exogenous component $\mathbf{V}$ and error term $\mathbf{D}$] For arbitrary $0 \leq t \leq \mathcal{T}$, we assume the following orthogonality condition:

$$[\mathbf{V}^\circ, \mathbf{D}^\circ]_t = 0, \quad (6.2)$$

$$[\mathbf{D}^\circ, \mathbf{U}^\circ]_t = 0, \quad (6.3)$$

$$[\mathbf{V}^\circ, \mathbf{Z}^\circ]_t = 0. \quad (6.4)$$

Based on the above assumption, it is easy to see that the correlation between observable factors and latent factors are determined by the correlation between $\mathbf{D}$ and $\mathbf{L}$. Note that although [Assumption 7](#) assumes that $\mathbf{V}_t$ is orthogonal to $\mathbf{Z}_t$, $\mathbf{D}_t$ can be correlated with $\mathbf{Z}_t$.

Based on [Eq. \(6.2\)](#) and the standard theory about high frequency regression, see, i.e., [Aït-Sahalia et al. \(2020\)](#) and [Chen et al. \(2023\)](#), we could find the OLS solution of δ_t as:

$$\delta_t^{\text{OLS}} = \left([\mathbf{V}^\circ, \mathbf{V}^\circ]'_t \right)^{-1} [\mathbf{V}^\circ, \mathbf{X}^\circ]'_t. \quad (6.5)$$

By further defining:

$$d\mathbf{X}_t^{\text{OLS}} = (\delta_t^{\text{OLS}})^\top d\mathbf{V}_t, \quad (6.6)$$

we have:

$$d\mathbf{Y}_t = \beta_t^\top d\mathbf{X}_t^{\text{OLS}} + d\tilde{\mathbf{Z}}_t, \quad (6.7)$$

with

$$d\tilde{\mathbf{Z}}_t = \beta_t^\top d\mathbf{D}_t + d\mathbf{Z}_t, \quad (6.8)$$

and we know that $\mathbf{X}_t^{\text{OLS}}$ is orthogonal to $\tilde{\mathbf{Z}}_t$. Based on the idea of classical two-step least squares method, we can get an unbiased estimator for β_t as follows:

$$\beta_t^{2\text{SLS}} = \left(\mathbf{M}_t^{\mathbf{X}, \mathbf{X}} \right)^{-1} \mathbf{M}_t^{\mathbf{X}, \mathbf{Y}},$$

where

$$\begin{aligned}\mathbf{M}_t^{\mathbf{X},\mathbf{X}} &= [\mathbf{X}^\circ, \mathbf{V}^\circ]'_t \left([\mathbf{V}^\circ, \mathbf{V}^\circ]'_t \right)^{-1} [\mathbf{V}^\circ, \mathbf{X}^\circ]'_t, \\ \mathbf{M}_t^{\mathbf{X},\mathbf{Y}} &= [\mathbf{X}^\circ, \mathbf{V}^\circ]'_t \left([\mathbf{V}^\circ, \mathbf{V}^\circ]'_t \right)^{-1} [\mathbf{V}^\circ, \mathbf{Y}^\circ]'_t.\end{aligned}$$

Finally, the quadratic covariation of $\tilde{\mathbf{Z}}^\circ$ can be expressed as:

$$[\tilde{\mathbf{Z}}^\circ, \tilde{\mathbf{Z}}^\circ]'_t = c_t^{\mathbf{Y},\mathbf{Y}} - \mathbf{M}_t^{\mathbf{Y},\mathbf{X}} \left(\mathbf{M}_t^{\mathbf{X},\mathbf{X}} \right)^{-1} \mathbf{M}_t^{\mathbf{X},\mathbf{Y}}.$$

6.2. High frequency unsupervised exogenous component learning

In practice, it is not easy to find proper exogenous component which satisfying Eq. (6.1) and Assumption 7. By incorporating the idea of principal component analysis, this paper proposes the high frequency unsupervised exogenous component learning (HF-UECL) to overcome this difficulty. In the proposed HF-UECL method, we proposed the singular value decomposition to estimate the exogenous component.

To derive the solution of HF-UECL, we suppose the exogenous component $\mathbf{V}_t$ to be a linear combination of $\mathbf{Y}_t$, i.e.,

$$\mathbf{dV}_t = \gamma_t^\top \mathbf{dY}_t, \quad (6.9)$$

where $\gamma_t = \left\{ \gamma_t^{(r,s)} \right\}_{1 \leq r \leq p, 1 \leq s \leq \bar{d}}$. Then it is straightforward to see that

$$[\mathbf{V}^\circ, \mathbf{X}^\circ]'_t = \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}}, \quad (6.10a)$$

$$[\mathbf{V}^\circ, \mathbf{Y}^\circ]'_t = \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}}, \quad (6.10b)$$

$$[\mathbf{V}^\circ, \mathbf{V}^\circ]'_t = \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t. \quad (6.10c)$$

Based on the orthogonality between $\mathbf{Z}_t$ and $\mathbf{V}_t$ (enabled by Assumption 7), we have:

$$[\mathbf{V}^\circ, \mathbf{Y}^\circ]'_t = [\mathbf{V}^\circ, \mathbf{X}^\circ]'_t \beta_t,$$

which implies that

$$\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} = \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \beta_t. \quad (6.11)$$

Based on Eq. (6.10a)–(6.10c), we have:

$$\mathbf{M}_t^{\mathbf{X},\mathbf{X}} = c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}},$$

$$\mathbf{M}_t^{\mathbf{X},\mathbf{Y}} = c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}},$$

and

$$\begin{aligned}[\tilde{\mathbf{Z}}^\circ, \tilde{\mathbf{Z}}^\circ]'_t &= c_t^{\mathbf{Y},\mathbf{Y}} - c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \left(c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \right)^{-1} c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}}.\end{aligned} \quad (6.12)$$

Moreover, the two-step least squares solution for β_t can be further expressed as:

$$\beta_t^{2SLS} = \left(c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \right)^{-1} c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}}. \quad (6.13)$$

Plugging Eq. (6.13) into Eq. (6.11), we obtain an important identity:

$$\gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \left(c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \left(\gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t \right)^{-1} \gamma_t^\top c_t^{\mathbf{Y},\mathbf{X}} \right)^{-1} c_t^{\mathbf{X},\mathbf{Y}} \gamma_t \equiv \gamma_t^\top c_t^{\mathbf{Y},\mathbf{Y}} \gamma_t. \quad (6.14)$$

Finally, the solution to γ_t is stated in the following theorem.

Theorem 2. [Solution to High Frequency Unsupervised Exogenous Component Learning] Under Assumptions 1–4 and Assumption 7, suppose $\text{rank}(c_t^{\mathbf{X},\mathbf{Y}}) \equiv d$ for all t , then the HF-UECL solution for γ_t can be expressed as:

$$\gamma_t^{UECL} = \left(c_t^{\mathbf{X},\mathbf{Y}} \right)^+.$$

This implies that $d = \bar{d}$.

Plugging the result of Theorem 2 into Eq. (6.13), we have:

$$\beta_t^{UECL} = \left(c_t^{\mathbf{Y},\mathbf{X}} \right)^+ c_t^{\mathbf{Y},\mathbf{Y}}. \quad (6.15)$$

Given the solutions γ_t^{UECL} and β_t^{UECL} , it is straightforward to verify that $[\mathbf{V}^\circ, \mathbf{Z}^\circ]_t = 0$ as imposed in [Assumption 7](#).

Finally, the estimate of β_t^{UECL} can be expressed by its sample analogue:

$$\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL}} = \left(\hat{c}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{X}} \right)^+ \hat{c}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{Y}}. \quad (6.16)$$

Remark 2 (Connection to Instrumental Variable Analysis). The proposed HF-UECL method has several similarities to the classic inference of instrumental variables (IV). For example, (i) the modeling approach of exogenous components, i.e., Eq. (6.1) and [Assumption 7](#), is similar to that of classical IV (more specifically, say, strong IV); (ii) the derivation of exogenous components employs the two-step least squares method, which is frequently used in classic IV inference. However, the differences between UECL and classic IV inference include but not limited to the following fact: the exogenous component are assumed to be unobservable in this paper.

Remark 3 (Brief Review on Projection-based Method). In low-frequency settings, the endogenous observable factors have been investigated based on the well-known projection-based methods, see, e.g., [Fan et al. \(2016b\)](#), [Kelly et al. \(2020\)](#), [Bryzgalova et al. \(2023\)](#). The main difference between the projection-based method and the one proposed in this paper is that the projection-based method models the observable covariates in the factor loadings, while the HF-UECL method places them into factors, i.e., in the parallel role to the latent factors. The motivation for the modeling approach in the projection-based method is as follows. The low-frequency factor model structure can be expressed as follows:

$$y_{i,t} = \beta_i X_{i,t} + \theta_i L_{i,t} + e_{i,t},$$

where X and L are the observable and latent factors, respectively; β and θ are the loadings to the observable and latent factors, respectively. Under low-frequency setting, the reason that observable factors can be modeled into the factor loadings can be illustrated by the following derivation:

$$y_{i,t} = \beta_i \times \frac{X_{i,t}}{L_{i,t}} \times L_{i,t} + \theta_i \times L_{i,t} + e_{i,t} \quad (6.17)$$

$$= \left(\theta_i + \beta_i \times \frac{X_{i,t}}{L_{i,t}} \right) L_{i,t} + e_{i,t}, \quad (6.18)$$

where the observable covariates become part of the factor loading. Note that in Eq. (6.18), the role of factor, i.e., L , is purely latent. This is the modeling foundation of the projection-based method under low frequency. However, such kind of modeling benefit is not applicable under high-frequency setting. For example, the observable and latent factors, i.e., $X_{i,t}$ and $L_{i,t}$, become the differential operators, $dX_t^{(i)}$ and $dL_t^{(i)}$ in the continuous-time model, and the ratio $\frac{dX_t^{(i)}}{dL_t^{(i)}}$ does not exist. This is the reason that we conclude that the UECL method proposed in this paper is a non-trivial extension of the projection-based method in mathematical essence. However, we should also emphasize that the projection-based method can simultaneously finish the goals in both [Sections 5](#) and [6](#) if we ignore its setting on observation frequency. This is the theoretical benefit of the projection-based method, and it will be interesting to explore its possible extension to high frequency observations.

Under finite dimensionality, i.e., p, d, q are fixed in asymptotics, $\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{HF-UECL}}$ is consistent, i.e.,

$$\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL}} = \beta_{T_{n,i-1}}^{\text{UECL}} + o_p(1),$$

provided the consistency of $\hat{c}_{\Delta T_n, T_{n,i-1}}$. Moreover, because β_t^{UECL} is a nonlinear functional of the spot covariance c_t , a bias corrected estimator for $\int_0^T \beta_t^{\text{UECL}} dt$ can be constructed by following the theory established in [Chen \(2024\)](#), its central limit theorem and asymptotic variance estimator can also be established based on the aforementioned theory.

6.3. Exogeneity test for observable factors

The statistical inference related to the proposed unsupervised exogenous components learning method can be further applied to testing the exogeneity of the observable factors. In this subsection, we omit the symbol for continuous part of a stochastic process, i.e., “ $\circ$ ”. Recall Eq. (2.4), i.e.,

$$d\mathbf{Y}_t = \beta_t^\top d\mathbf{X}_t + \theta_t^\top d\mathbf{L}_t + d\mathbf{U}_t.$$

When $\mathbf{X}$ and $\mathbf{L}$ are correlated, it is usually interesting to investigate the projection of the latent factors (e.g., $\mathbf{L}$) on the direction of observable factors (e.g., $\mathbf{X}$). Define

$$d\mathbf{H}_t = \theta_t^\top d\mathbf{L}_t,$$

then the basic inference on latent factors can be derived by setting the following regression:

$$d\mathbf{H}_t = \mathbf{b}_t^\top d\mathbf{X}_t + d\mathbf{E}_t, \quad (6.19)$$

where we suppose $\langle \mathbf{X}, \mathbf{E} \rangle_t' \equiv 0$ and $\langle \mathbf{U}, \mathbf{E} \rangle_t' \equiv 0$. We know that

$$d\mathbf{Y}_t = (\beta_t + \mathbf{b}_t)^\top d\mathbf{X}_t + d\tilde{\mathbf{U}}_t \text{ with } d\tilde{\mathbf{U}}_t = d\mathbf{E}_t + d\mathbf{U}_t \quad (6.20)$$

and

$$\langle \mathbf{X}, \tilde{\mathbf{U}} \rangle_t' = 0.$$

Based on the theory of realized regression, the OLS estimator for $\beta_t + \mathbf{b}_t$ is $(c_t^{\mathbf{X},\mathbf{X}})^{-1} c_t^{\mathbf{X},\mathbf{Y}}$. Plug in Eq. (6.15), we obtain:

$$\mathbf{b}_t = (c_t^{\mathbf{X},\mathbf{X}})^{-1} c_t^{\mathbf{X},\mathbf{Y}} - (c_t^{\mathbf{Y},\mathbf{X}})^+ c_t^{\mathbf{Y},\mathbf{Y}},$$

where $(c_t^{\mathbf{Y},\mathbf{X}})^+$ denotes the Moore–Penrose inverse of $c_t^{\mathbf{Y},\mathbf{X}}$.

Definition 2 (Exogeneity Test for Observable Factors). For observable factors $(\mathbf{X}_t)$, the exogeneity test can be defined as follows:

$$H_0 : \int_0^T \mathbf{b}_t dt = 0 \text{ versus } H_1 : \int_0^T \mathbf{b}_t dt \neq 0. \quad (6.21)$$

If we fail to reject $\int_0^T \mathbf{b}_t dt = 0$ significantly, it means that the latent factors and observable factors are statistically uncorrelated; This means that the inference about $\int_0^T \mathbf{b}_t dt$ will eventually result in a hypothesis test about the orthogonality between latent factors and observable factors.

The model structure in Eq. (6.19) also provides a way to understand the endogenous observable factors. More specifically, if the observable factors $(\mathbf{X}_t)$ is exogenous with respect to $(\int_0^t \theta_s^T d\mathbf{L}_s)$, then the approach proposed in Section 5 is still useful. In this case, the problem of correlated latent and observable factors has eventually been reduced to the orthogonal case. Then the theories established in Theorem 1 can be directly applied to detect if there is any significant components existing in the unexplainable part, e.g. $\tilde{\mathbf{U}}_t$.

6.4. Theoretical result for $\int_0^T \beta_t^{\text{UECL}} dt$ estimation

The parameter which we would like to estimate is:

$$\mathfrak{D}_T \triangleq \int_0^T \beta_t^{\text{UECL}} dt.$$

The bias-corrected estimator for $\mathfrak{D}_T$ can be constructed as follows:

$$\tilde{\mathfrak{D}}_T \triangleq \Delta T_n \sum_{i=1}^{B_n} [\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL}} - \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL-BC}}], \quad (6.22)$$

where $\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL}}$ is defined in Eq. (6.16), and the bias-correction term $\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL-BC}}$ is defined as:

$$\begin{aligned} \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL-BC}} &= 2 \left(\hat{c}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{X}} \right)^+ \hat{\mathfrak{C}}_{T_{n,i-1}}^{(1)} \hat{\phi}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{Y}} - 2 \left(\hat{c}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{X}} \right)^+ \hat{\mathfrak{C}}_{T_{n,i-1}}^{(2)} \hat{c}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{Y}} \\ &\quad - \mathbf{1}_{p+d}^T \hat{\phi}_{T_{n,i-1}} \mathbf{1}_{p+d} \left(\hat{c}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{X}} \right)^+ \left(\hat{c}_{T_{n,i-1}}^{\mathbf{X},\mathbf{Y}} \right)^+ \hat{\mathfrak{C}}_{T_{n,i-1}}^{(3)} \hat{c}_{T_{n,i-1}}^{\mathbf{Y},\mathbf{Y}}, \end{aligned} \quad (6.23)$$

with $\hat{\mathfrak{C}}_{T_{n,i-1}}^{(1)}$, $\hat{\mathfrak{C}}_{T_{n,i-1}}^{(2)}$, and $\hat{\mathfrak{C}}_{T_{n,i-1}}^{(3)}$ being defined in Eqn. (E.12a)–(E.12c) of Appendix E.2.

For the asymptotic variance estimator of $\tilde{\mathfrak{D}}_T^{(r,s)}$, we construct it as follows:

$$\widehat{\text{AVAR}} \left(\tilde{\mathfrak{D}}_T^{(r,s)} \right) = \Delta T_n^2 \sum_{i=1}^{B_n} \left[\hat{\mathfrak{A}}_{T_{n,i-1}}^{(r,s)} \right]^2, \quad (6.24)$$

where $\hat{\mathfrak{A}}_{T_{n,i-1}}^{(r,s)}$ is defined in Eqn. (E.13) of Appendix E.2.

Finally, we state the central limit theorem for $\tilde{\mathfrak{D}}_T^{(r,s)}$ as follows.

Theorem 3. Suppose all assumptions in Theorem 2 hold. Suppose Assumptions 1–4 and further assume $a_n^{-1} \Delta T_n \rightarrow 0$ and $a_n^{-3/2} \Delta T_n \rightarrow \infty$ as $n \rightarrow \infty$. For the setting of Truncated S-TSRV, suppose Assumptions B.1–B.3 as stated in Appendix B. Suppose d and p are finite in asymptotics. Then for $1 \leq r \leq d$, $1 \leq s \leq p$, we obtain:

$$\frac{\tilde{\mathfrak{D}}_T^{(r,s)} - \mathfrak{D}_T^{(r,s)}}{\sqrt{\widehat{\text{AVAR}} \left(\tilde{\mathfrak{D}}_T^{(r,s)} \right)}} \xrightarrow{\mathcal{L}} N(0, 1)$$

stably.

The proof of theorem mainly relies on the Lemmas L.1–L.2 in Appendix L and Proposition 2 of Chen (2024).

6.5. Theoretical result for $\int_0^T \mathbf{b}_t dt$ estimation

The parameter which we would like to estimate is:

$$\mathfrak{L}_T \triangleq \int_0^T \mathbf{b}_t dt.$$

The bias-corrected estimator for $\mathfrak{L}_T$ can be constructed as follows:

$$\tilde{\mathfrak{L}}_T \triangleq \Delta T_n \sum_{i=1}^{B_n} \left[\hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}} - \hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}}^{\text{BC}} \right], \quad (6.25)$$

where $\hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}}$ is defined as

$$\hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}} = \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{OLS}} - \left(\hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{X}} \right)^+ \hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{Y}, \mathbf{Y}}$$

and the bias-correction term $\hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}}^{\text{BC}}$ is defined as:

$$\hat{\mathbf{b}}_{\Delta T_n, T_{n,i-1}}^{\text{BC}} = \left(\mathbf{b}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \right)^2 \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{OLS}} - \mathbf{b}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \left(\hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \right)^{-1} \hat{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{Y}} - \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL-BC}}$$

with

$$\begin{aligned} \hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{OLS}} &\triangleq \left(\hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \right)^{-1} \hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{Y}}, \\ \mathbf{b}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} &\triangleq \left(\hat{\mathbf{c}}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}} \right)^{-1} \hat{\phi}_{\Delta T_n, T_{n,i-1}}^{\mathbf{X}, \mathbf{X}}, \end{aligned}$$

and $\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{UECL-BC}}$ being defined in Eq. (6.23). Although $\hat{\beta}_{\Delta T_n, T_{n,i-1}}^{\text{OLS}}$ has already been defined in Eq. (5.11), we add the superscript “OLS” in order to emphasize that it is the estimator for the OLS beta, rather than the UECL beta.

For the asymptotic variance estimator of $\tilde{\mathfrak{L}}_T^{(r,s)}$, we construct it as follows:

$$\widehat{\text{AVAR}} \left(\tilde{\mathfrak{L}}_T^{(r,s)} \right) = \Delta T_n^2 \sum_{i=1}^{B_n} \left[\hat{\mathfrak{L}}_{\Delta T_n, T_{n,i-1}}^{(r,s)} \right]^2, \quad (6.26)$$

where $\hat{\mathfrak{L}}_{\Delta T_n, T_{n,i-1}}^{(r,s)}$ is defined in Eqn. (E.14) of Appendix E.2.

Finally, we state the central limit theorem for $\tilde{\mathfrak{L}}_T^{(r,s)}$ as follows.

Theorem 4. Suppose all assumptions in Theorem 2 hold. Suppose Assumptions 1–4 and further assume $a_n^{-1} \Delta T_n \rightarrow 0$ and $a_n^{-3/2} \Delta T_n \rightarrow \infty$ as $n \rightarrow \infty$. For the setting of Truncated S-TSRV, suppose Assumptions B.1– B.3 as stated in Appendix B. Suppose d and p are finite in asymptotics. Then for $1 \leq r \leq d$, $1 \leq s \leq p$, we obtain:

$$\frac{\tilde{\mathfrak{L}}_T^{(r,s)} - \mathfrak{L}_T^{(r,s)}}{\sqrt{\widehat{\text{AVAR}} \left(\tilde{\mathfrak{L}}_T^{(r,s)} \right)}} \xrightarrow{\mathcal{L}} N(0, 1)$$

stably.

The proof of theorem mainly relies on the proof of Theorem 3 and the Section B.2 of Aït-Sahalia et al. (2020).

7. Monte Carlo simulation

In the Monte Carlo experiment, the data is generated from the model defined in Eq. (2.4), i.e.,

$$d\mathbf{Y}_t = \hat{\rho}_t^\top d\mathbf{X}_t + \theta_t^\top d\mathbf{L}_t + d\mathbf{U}_t.$$

Before specifying the settings of the exogenous case and endogenous case, we first define several building blocks:

$$\begin{aligned} d\mathcal{X}_t^{(i)} &= \mu_i dt + \sigma_i^{(i)} d\mathcal{W}_t^{(i)} + dJ_t^{\mathcal{X}, (i)} \\ d\mathcal{L}_t^{(j)} &= \check{\mu}_j dt + \check{\sigma}_j^{(j)} d\check{\mathcal{W}}_t^{(j)} + dJ_t^{\mathcal{L}, (j)} \\ d\mathbf{U}_t^{(k)} &= v_k d\mathcal{B}_t^{(k)} + dJ_t^{\mathbf{U}, (k)}, \end{aligned}$$

where $i = 1, 2, \dots, d$, $j = 1, 2, \dots, q$ and $k = 1, 2, \dots, p$. Here we set $d = q = 3$. The jump processes $\left\{ \left(J_t^{\mathcal{X}, (i)} \right)_{0 \leq t \leq T} \right\}_{1 \leq i \leq d}$, $\left\{ \left(J_t^{\mathcal{L}, (j)} \right)_{0 \leq t \leq T} \right\}_{1 \leq j \leq q}$ and $\left\{ \left(J_t^{\mathbf{U}, (k)} \right)_{0 \leq t \leq T} \right\}_{1 \leq k \leq p}$ are driven by three Poisson processes with arrival rates $\lambda^{\mathcal{X}}$, $\lambda^{\mathcal{L}}$ and $\lambda^{\mathbf{U}}$, respectively. Their jump sizes follow double exponential distributions with means denoted by $\mu_+^{\mathcal{X}}, \mu_-^{\mathcal{X}}, \mu_+^{\mathcal{L}}, \mu_-^{\mathcal{L}}, \mu_+^{\mathbf{U}}$ and $\mu_-^{\mathbf{U}}$, respectively. We set $\lambda^{\mathcal{X}} = \lambda^{\mathcal{L}} = \lambda^{\mathbf{F}}$, $\mu_+^{\mathcal{X}} = \mu_+^{\mathcal{L}} = \mu_+^{\mathbf{F}}$, and $\mu_-^{\mathcal{X}} = \mu_-^{\mathcal{L}} = \mu_-^{\mathbf{F}}$.

We simulate the factor loading in the following scheme:

$$d\beta_t^{(j,i)} = \begin{cases} \tilde{\kappa}_1 \left(\tilde{\theta}_{i1} - \beta_t^{(j,i)} \right) dt + \tilde{\xi}_1 \sqrt{\beta_t^{(1,i)}} d\tilde{B}_t^{(1,i)} & \text{if } j = 1, \\ \tilde{\kappa}_j \left(\tilde{\theta}_{ij} - \beta_t^{(j,i)} \right) dt + \tilde{\xi}_j d\tilde{B}_t^{(j,i)} & \text{if } j \geq 2. \end{cases}$$

and

$$d\mathbf{b}_t^{(j,i)} = \begin{cases} \dot{\kappa}_1 \left(\dot{\theta}_{i1} - \mathbf{b}_t^{(1,i)} \right) dt + \dot{\xi}_1 \sqrt{\mathbf{b}_t^{(1,i)}} d\dot{B}_t^{(1,i)} & \text{if } j = 1, \\ \dot{\kappa}_j \left(\dot{\theta}_{ij} - \mathbf{b}_t^{(j,i)} \right) dt + \dot{\xi}_j d\dot{B}_t^{(j,i)} & \text{if } j \geq 2. \end{cases}$$

and

$$d\theta_t^{(r,s)} = \check{\kappa}_j \left(\check{\theta}_{ij} - \theta_t^{(r,s)} \right) dt + \check{\xi}_j d\check{B}_t^{(r,s)} \text{ for } 1 \leq r \leq q, 1 \leq s \leq p.$$

In this simulation, the first component of $\mathbf{X}$ is set as the market factor. Thus, its factor loadings $\beta_{\bullet,1}$ and $\mathbf{b}_{\bullet,1}$ are positive.

The correlation matrices of $d\mathcal{W}$ and $d\check{\mathcal{W}}$ are defined as $\rho^{\mathcal{X}}$ and $\rho^{\mathcal{L}}$. The volatility processes of $\mathcal{X}$, $\mathcal{L}$ and $\mathbf{U}$ are simulated as follows:

$$\begin{aligned} d\left(\sigma_t^{(j)}\right)^2 &= \kappa_j \left(\theta_j - \left(\sigma_t^{(j)}\right)^2 \right) dt + \eta_j \sigma_t^{(j)} d\tilde{\mathcal{W}}_t^{(j)}, \\ d\left(\check{\sigma}_t^{(j)}\right)^2 &= \bar{\kappa}_j \left(\bar{\theta}_j - \left(\check{\sigma}_t^{(j)}\right)^2 \right) dt + \bar{\eta}_j \check{\sigma}_t^{(j)} d\check{\mathcal{W}}_t^{(j)}, \\ dv_t^2 &= \kappa \left(\theta - v_t^2 \right) dt + \eta v_t d\tilde{B}_t, \end{aligned}$$

where the correlation between $d\mathcal{W}^{(j)}$ and $d\check{\mathcal{W}}^{(j)}$ is ρ_j and the correlation between $d\check{\mathcal{W}}^{(j)}$ and $d\tilde{\mathcal{W}}^{(j)}$ is $\check{\rho}_j$.

Recall the general model settings in Eq. (2.4), we specify the simulation scheme as follows:

1. Exogenous Case: we set:

$$\mathbf{X}_t \equiv \mathcal{X}_t, \mathbf{L}_t \equiv \mathcal{L}_t, \text{ and } \theta_t \equiv \vartheta_t,$$

2. Endogenous Case: we set:

$$\mathbf{X}_t \equiv \mathcal{X}_t, \text{ and } \theta_t^\top d\mathbf{L}_t \equiv \mathbf{b}_t^\top d\mathcal{X}_t + \vartheta_t^\top d\mathcal{L}_t.$$

The parameters used in the simulation are reported in Table D.1 of Appendix D.

The processes are sampled in the equidistant grid with $\Delta t_n = 1$ second. And the observed processes are contaminated by microstructure noise:

$$\Theta_{t_j}^* = \Theta_{t_j} + \epsilon_{t_j}, \quad (7.1)$$

where ϵ_{t_j} are i.i.d. $(p+d)$ -dimensional random vectors, sampled from $\mathcal{N}_{p+d}(0, \Sigma^\epsilon)$, with $\Sigma^\epsilon = \Phi\Phi^\top$ and $\Phi = (\Phi_1, \Phi_2, \dots, \Phi_{p+d})^\top$. Note that $\Phi_1, \Phi_2, \dots, \Phi_{p+d}$ are i.i.d. random variables from $N(0, (0.0005)^2)$. The time horizon in the simulation experiment is set as: $\mathcal{T} = 1$ week (assume 1 week consists of 5 trading days and the length of each trading day is 1/252). We assume that a trading day consists of 6.5 h for open trading.

7.1. Simulation results

In the case of exogenous setting, we apply the bias-corrected estimator $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ (as defined in Eq. (5.19)) to estimate the integrated eigenvalues of the residual covariance, i.e., $\mathfrak{V}_{\mathcal{T}}^{(l)}$ (as defined in Eq. (5.17)–(5.18)). In the simulation, the observations are contaminated by noise and jumps. The computation is conducted under different combinations of stocks number and sampling frequency. For the parameter u_n in truncation method, the standard procedure in the literature is employed, i.e., $u_n^{(r)} = \max(u_n^{(r),+}, u_n^{(r),-})$,

$u_n^{(r),+} = \left(2E[\widehat{\epsilon^{(r)}}]^2 / \bar{\mathcal{M}}_n^{(r)} \right)^{0.47}$, $u_n^{(r),-} = 3 \left(\frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} c_t^{(r,r)} dt \right)^{1/2} \Delta\tau_n^{0.47}$, where $\widehat{E[\epsilon^{(r)}]}^2 = \frac{1}{2n^{(r)}} \sum_{j=1}^{n^{(r)}} \left(\theta_{t_j^{(r)}}^{*,(r)} - \theta_{t_{j-1}^{(r)}}^{*,(r)} \right)^2$, $\bar{\mathcal{M}}_n^{(r)} = N^{-1} \sum_{i=1}^N \mathcal{M}_{n,i}^{(r)}$ and $\int_0^{\mathcal{T}} c_t^{(r,r)} dt$ can be estimated by bipower variations with sampling interval of 5 min. The number of stocks $p = 5, 10, 20, 30, 50$ and 80, while the sampling frequency is set in three scenarios:

1. $\Delta\tau_n = 5$ seconds and $\Delta T_n = 2340\Delta\tau_n$, with $K = 20, J = 10$.
2. $\Delta\tau_n = 15$ seconds and $\Delta T_n = 600\Delta\tau_n$, with $K = 10, J = 5$.
3. $\Delta\tau_n = 1$ minute and $\Delta T_n = 150\Delta\tau_n$, with $K = 4, J = 2$.

All tables and figures generated in this section are collected in Appendix D.

Table D.2 reports the results based on Truncated S-TSRV. As a comparison, Table D.4 reports the results based on Simple CV. For each table, the second column reports the averaged true values in the simulation; the 3rd-5th columns report the summary statistics of the estimation errors and standard error estimates when $\Delta\tau_n = 5$ seconds; the 6th-8th columns report the summary statistics when $\Delta\tau_n = 15$ seconds; the 9th-11th columns report the summary statistics when $\Delta\tau_n = 1$ minute.

The simulation result in Table D.2 suggests that, under all combinations of stock number and sampling frequency, the Truncated S-TSRV based estimator $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ can achieve satisfactory performance by accurately recovering the true integrated eigenvalues $\mathfrak{V}_{\mathcal{T}}^{(l)}$, $l = 1, 2, 3$ and 4. For each $l = 1, 2, 3$ and 4, the larger number of stocks yields the larger value of the averaged integrated eigenvalues.

When increasing the length of sampling interval, the absolute mean and the standard deviation of the estimation error tend to be magnified. Moreover, the 4th-5th, 7th-8th and 10th-11th columns of the table show that the standard error estimator defined in Eq. (5.23) can capture the main variation in the estimation error when $l = 1, 2, 3$ and 4.

Note that the accuracy of $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ is benefited from the Truncated S-TSRV estimator, which is robust to jumps, microstructure noise and asynchronous observation times. As a comparison, the estimation performance of the parallel estimator implemented by the Simple CV is reported in Table D.4, which shows that Simple CV based estimator is heavily biased in the presence of microstructure noise. For example, when $\Delta_n = 5, 15$ or 60 seconds, $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ (as defined in Eq. (K.1)) tends to over-estimate the true eigenvalue $\mathfrak{V}_{\mathcal{T}}^{(l)}$ for $l = 1, 2, 3$ and 4. For simple eigenvalues (i.e., $l = 1, 2, 3$) and fixed Δ_n , the absolute value of the estimation bias of $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ tends to be magnified as the cross-sectional dimension p increasing. Moreover, the standard error estimator (as proposed in Eq. (K.2)) cannot work when the observations are noisy. For example, for all choices of p, d and Δ_n , the standard error estimator tends to under-estimate the square root of the asymptotic variance of $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$.

To show the asymptotic normality of the bias corrected estimator $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$, the finite sample distribution of the standardized statistics for $p = 30$ stocks are reported in Figure D.1, where $\Delta\tau_n = 5$ seconds. Note that the standardized statistics are calculated by the following formulas:

$$\frac{\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)} - \mathfrak{V}_{\mathcal{T}}^{(l)}}{\widehat{\text{AVAR}}^{\frac{1}{2}}},$$

for the standardized statistics of bias-corrected estimator, while

$$\frac{\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)} - \mathfrak{V}_{\mathcal{T}}^{(l)}}{\widehat{\text{AVAR}}^{\frac{1}{2}}},$$

for the standardized statistics of the estimator before bias correction, where $\widehat{\text{AVAR}}$ is defined in Eq. (D.1), $\hat{\mathfrak{V}}_{\mathcal{T}}^{(l)}$ is defined in Eq. (G.1), and $\mathfrak{V}_{\mathcal{T}}^{(l)}$ is defined in Eq. (5.23).

In the case of endogenous setting, we implement the estimation based on the theories of $\tilde{\mathfrak{V}}_{\mathcal{T}}^{(r,s)}$ and $\tilde{\mathfrak{L}}_{\mathcal{T}}^{(r,s)}$, i.e., Theorems 3 and 4, respectively. In the simulation, we only use the sampling frequency of $\Delta\tau_n = 5$ seconds. Various choices of p has been considered, i.e., $p = 5, 10, 20, 30, 50$, and 80. The summary statistics for $\tilde{\mathfrak{V}}_{\mathcal{T}}^{(r,s)}$ and $\tilde{\mathfrak{L}}_{\mathcal{T}}^{(r,s)}$ are reported in Table D.3, which demonstrating the consistency of the estimators and the consistency of the asymptotic variance estimators. The asymptotic normality of the estimators has also been validated by histograms in Figure D.2 with $p = 80$ and $\Delta\tau_n = 5$ seconds.

8. Empirical study

As an empirical study, we apply our estimation methodology to the intraday returns of the S&P 100 Index (OEX) constituents. The stock prices are extracted from the Trade and Quote (TAQ) database of the New York Stock Exchange (NYSE). We collect the intraday stock prices of 80 most actively traded stocks among the S&P 100 Index constituents, between 9:35 a.m. EST and 3:55 p.m. EST of each trading day, ranging from January 2007 to December 2017 (2769 trading days in total). For the independent variables, we use the high frequency factors constructed by Aleti (2022). More specifically, we use the high frequency version of the Fama–French 5 + 1 factors, including MKT, SMB, HML, RMW, CMA, and UMD. The sampling frequency of this study is 1 min.

Therefore, in this model, we include the high frequency Fama–French 5 + 1 factors as the independent variables. The dependent variable $\mathbf{Y}_t$ will be the 80 selected components of S&P 100 Index. For example, the factor model can be expressed as:

$$\begin{aligned} d\mathbf{Y}_t^{\circ} &= \beta_t^{\top} d\mathbf{X}_t^{\circ} + d\mathbf{Z}_t^{\circ} \\ &= \beta_t^{\top} d\mathbf{X}_t^{\circ} + \theta_t^{\top} d\mathbf{L}_t^{\circ} + d\mathbf{U}_t^{\circ}, \end{aligned}$$

and the vector of observable factors $\mathbf{X}_t$ can be expressed as follows:

$$\mathbf{X} = (X^{(1)}, \dots, X^{(5)}) = (X^{\text{MKT}}, X^{\text{SMB}}, X^{\text{HML}}, X^{\text{RMW}}, X^{\text{CMA}}, X^{\text{UMD}}).$$

8.1. Regular inference on proposed estimates

In the exogenous case, i.e., under Assumption 5, we estimate the time-varying sequence of the integrated eigenvalues of the residual covariance matrix, and conduct the hypothesis tests for these estimates. The analysis results are plotted in Fig. 8.1. Fig. 8.1-(a) shows the proportion of the residual total variation explained by the residual principal components 1–5. Fig. 8.1-(b) shows the time series of the p -value generated from tests of the integrated eigenvalue estimate tests. Fig. 8.1-(c) shows the proportion of the total variation explained by the proposed partially observable factor models. Fig. 8.1 shows that (i) after Jan 2014, the proportion of residual total variation explained by residual PC1 increases from 15% to 35% in average; (ii) at the significance level of 5%, the variance of the residual PC1 becomes significant after Jan 2014; (iii) the first residual principal component contributes to the largest increments (13%) in the explained total variation by the partially observable factor model.

In the endogenous case, i.e., [Assumption 5](#) no longer holds, we estimate $\int_0^T \beta_t^{\text{UECL}} dt$ and $\int_0^T \mathbf{b}_t dt$ based on [Theorems 3](#) and [4](#), respectively. For each trading day, we calculate the normalized statistics:

$$\frac{\tilde{\Sigma}_T^{(r,s)} - \Sigma_T^{(r,s)}}{\sqrt{\widehat{\text{AVAR}}(\tilde{\Sigma}_T^{(r,s)})}}, \text{ and } \frac{\tilde{\Sigma}_T^{(r,s)} - \Sigma_T^{(r,s)}}{\sqrt{\widehat{\text{AVAR}}(\tilde{\Sigma}_T^{(r,s)})}},$$

and denote the corresponding p-values as $p_{\Sigma}^{(r,s)}$ and $p_{\Sigma}^{(r,s)}$, respectively. For r th regressor, we calculate the rate of significant loading estimate as follows:

$$\frac{1}{p} \sum_{s=1}^p p_{\Sigma}^{(r,s)}, \text{ and } \frac{1}{p} \sum_{s=1}^p p_{\Sigma}^{(r,s)},$$

of which the time series has been plotted in [Figs. 8.2](#). For the estimates of β^{UECL} , the significant rate of Market factor is relatively high (approximately 80%). The significant rates of the other 5 factors are relatively low (approximately 40%). For the estimates of $\mathbf{b}$, the significant rate of all regressors has a downward trend (in average from 35% to 10%) during 2007 and 2009, and starting from January 2010, their significant rates tend to move together (with an averaged level of 20%). To show the profitability of the residual PCs 1 and 5, we plot the time series of the Sharpe ratios of residual PCs 1 and 5 along with those of FF5+1 factors in [Fig. 8.4](#), which shows that the newly identified residual PCs have the comparable Sharpe ratios compared to observable factors, i.e., FF5+1 factors.

8.2. Anomaly tests

In this section, we conduct anomaly tests for the significant residual PCs and significant unsupervised exogenous components to see whether the FF5+1 model can explain the variations in these new testing assets. The time series of the p-values for the α estimates are reported in [Fig. 8.3](#), where the sub-figure (a) reports the p-values for residual PCs 1 and 5, and the sub-figure (b) reports the p-values for the unsupervised exogenous components (UEC) related to SMB, HML, and RMW factors. [Fig. 8.3](#) shows that

- the residual PC 1 during Jan 2016–Dec 2017,
- the residual PC 5 during Jan 2012–July 2013, Mar–May, 2016,
- the UEC corresponding to SMB during Jan 2009–Aug 2010, and Oct 2015–Dec 2017;
- the UEC corresponding to HML during Jan 2009–Aug 2010;
- the UEC corresponding to RMW during Jan 2009–Aug 2010, and Oct 2012–Sep 2013,

yield significant alphas at the significance level of 0.05.

8.3. Application in minimum variance portfolio construction

The benefit of the partially observed factor model is also corroborated by applying it in the construction of minimum variance portfolios. The solution to the weights of the minimum variance portfolio can be expressed as follows:

$$\omega^* = \frac{\Sigma^{-1} \mathbf{1}_d}{\mathbf{1}_d^\top \Sigma^{-1} \mathbf{1}_d}, \quad (8.1)$$

of which the detailed derivation could be found in [Ding et al. \(2021\)](#).

Factor model structure has been widely applied into the estimation of high dimensional covariance/precision matrices, see, e.g., [Ait-Sahalia and Xiu \(2017\)](#), [Kong \(2017\)](#), and [Chen et al. \(2020, 2023\)](#). The estimation theory of high dimensional covariance and precision matrices has also been extended to the proposed high frequency partially observable factor models. Due to the length limitation of the paper, we have included the estimation methodology, theoretical results and detailed derivation procedure into Appendix C.

Denote the covariance matrix at time period t by Σ_t and denote the covariance matrix estimate at time period t by $\hat{\Sigma}_t$. Then the construction of minimum variance portfolio relies on the following assumption that:

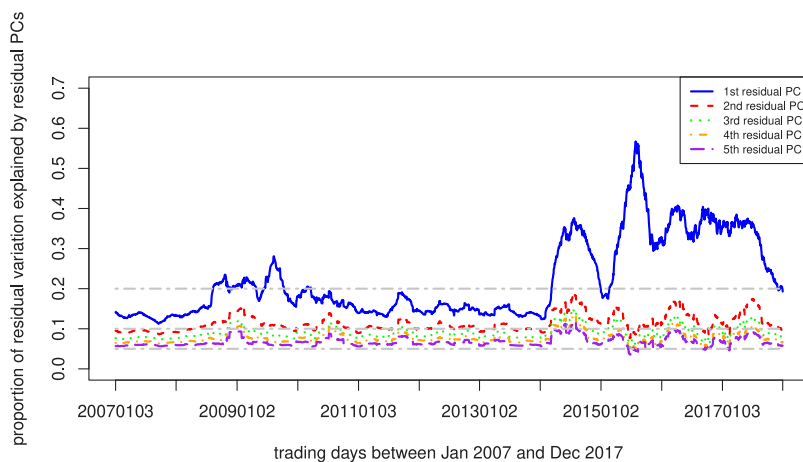
$$\Sigma_{t-1} = E(\Sigma_t | F_{t-1}). \quad (8.2)$$

Based on above assumption, the minimum variance portfolio is constructed in the way out-of-sample. For example the covariance matrix estimate on time period $t-1$ will be used to calculate the portfolio weights on time period t , i.e.,

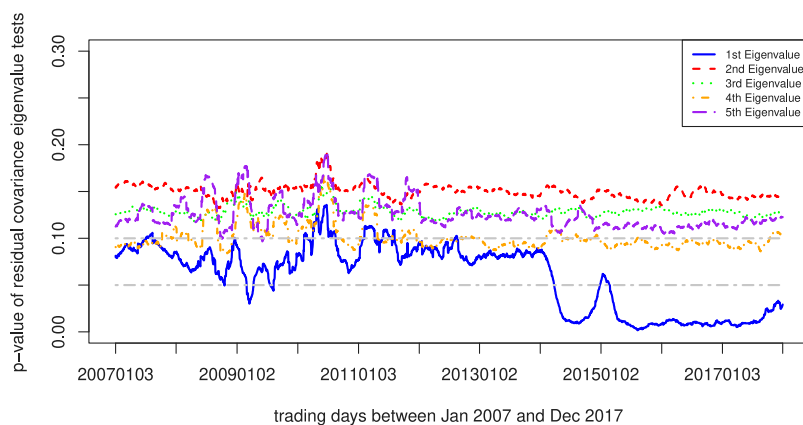
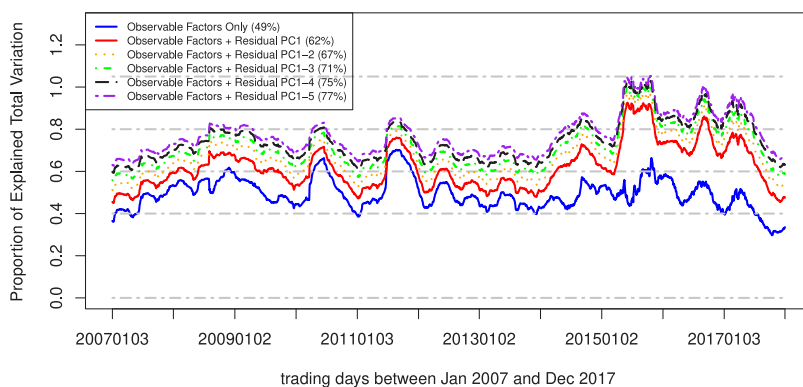
$$\omega_t^* = \frac{\hat{\Sigma}_{t-1}^{-1} \mathbf{1}_d}{\mathbf{1}_d^\top \hat{\Sigma}_{t-1}^{-1} \mathbf{1}_d}. \quad (8.3)$$

Suppose the cross-sectional daily return vector is denoted by $r_t = (r_t^{(1)}, r_t^{(2)}, \dots, r_t^{(p)})^\top$ for $t = 1, 2, \dots, T$, the MVP return r_t^{MVP} on t th trading day can be expressed as:

$$r_t^{\text{MVP}} = (\omega_t^*)^\top r_t. \quad (8.4)$$

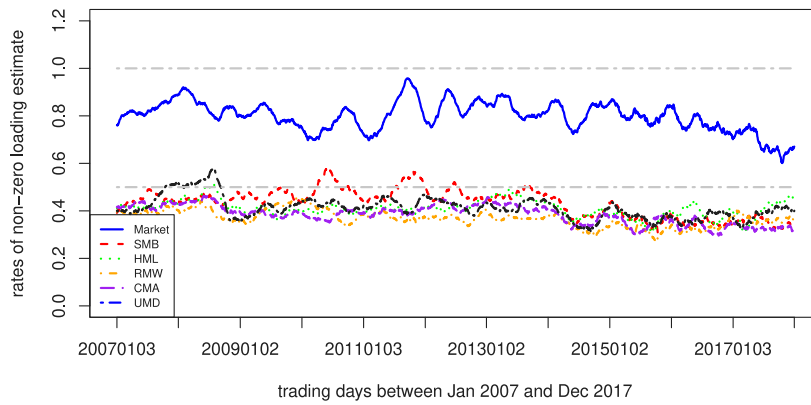
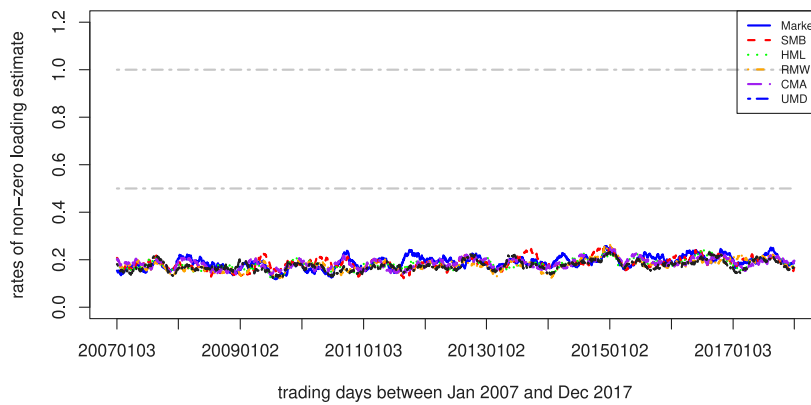


(a) Proportion of Residual Total Variation Explained by Residual PCs

(b) Time Series of the p -values from Residual PC Variance Test

(c) Proportion of Total Variation Explained by Observable Factors + Residual PCs

Fig. 8.1. Empirical results on residual principal components.

(a) Rate of Significant $\tilde{\mathcal{D}}_{\mathcal{T}}$ (b) Rate of Significant $\tilde{\mathcal{L}}_{\mathcal{T}}$ **Fig. 8.2.** Rates of non-zero loadings in endogenous case.

Notes. This figure reports the rates of significant estimates of $\tilde{\mathcal{D}}_{\mathcal{T}}^{(i,-)}$ and $\tilde{\mathcal{L}}_{\mathcal{T}}^{(i,-)}$, i.e., for $i = 1, 2, 3, 4, 5$ and 6, as defined in Eqs. (6.22) and (6.25), respectively. The significant level is set as 0.05.

The out-of-sample variance of the MVP is calculated as the sample variance of $\hat{r}_t^{\text{MVP}}$. The time frequency for t is daily in this study. We consider two models to construct the high dimensional covariance matrix:

- Model (I). The factor model only includes observable factors;
- Model (II). The factor model includes both observable factors and residual PC 1.

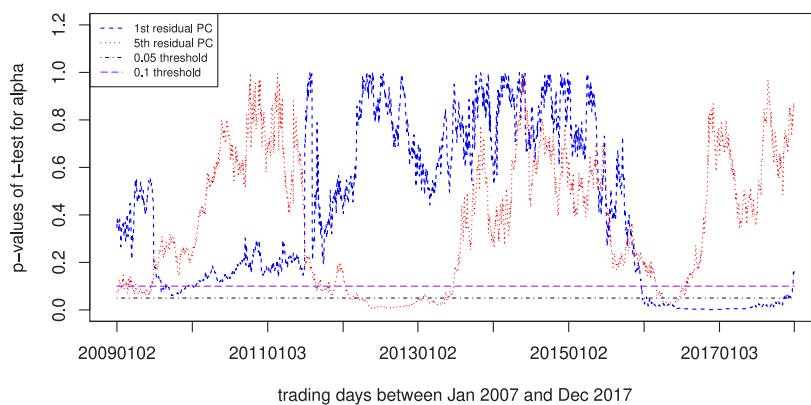
The Fig. 8.5 shows the risk comparison of the minimum variance portfolios constructed by different models, i.e., Model (I) and Model (II). It is evident that the high dimensional precision matrix estimated from partially observable factor model, i.e., Model (II), tends to obtain lower variance than that from purely observable factor models, which demonstrates the superior performance of the proposed model.

9. Conclusion

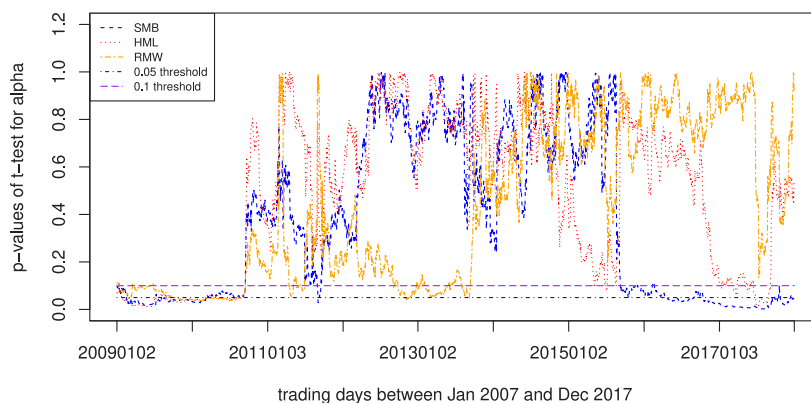
This paper mainly considers a novel model in high-frequency setting – partially observable factor model – which can simultaneously keep the economic interpretation and the higher explainability in second-order moment.

Theoretically, we proposed the estimation methodologies for the proposed model under the cases of both exogenous observable factors and endogenous observable factors. The high frequency unsupervised exogenous component learning (HF-UECL) method designed for the endogenous case can serve as a non-trivial extension of the Projected PCA or Instrumented PCA in low frequency setting.

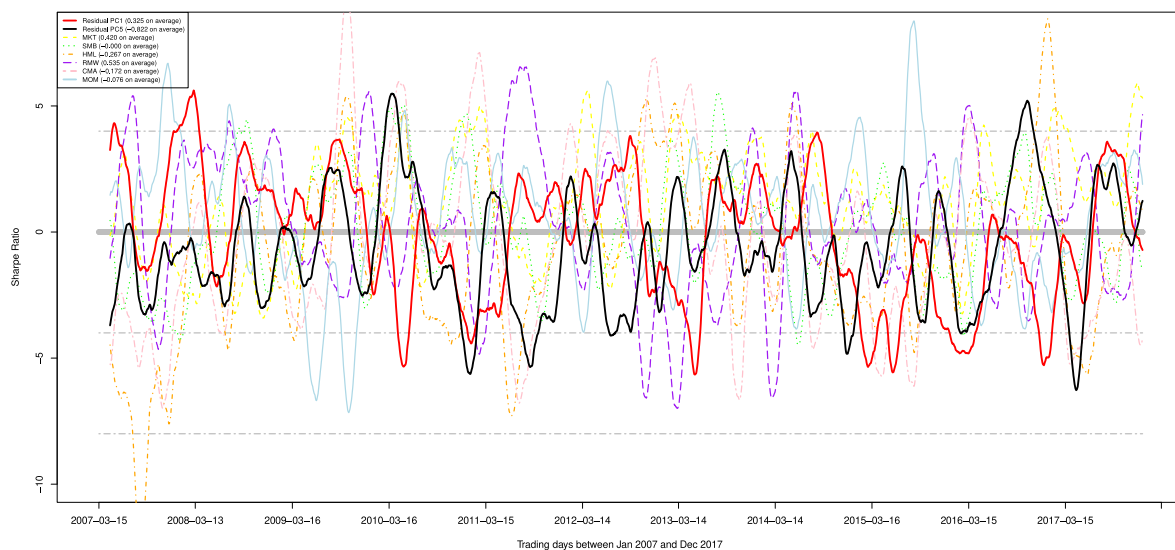
Empirically, by employing the high frequency version of the Fama–French 5+1 factors, we have identified one significant residual principal component, i.e., the residual PC 1. Moreover, as shown in the empirical study of minimum variance portfolio construction,



(a) Significant Residual Principal Components



(b) Significant Unsupervised Exogenous Component

Fig. 8.3. Anomaly tests for residual principal components and unsupervised exogenous components.**Fig. 8.4.** Time series of sharpe ratios of FF5+1 factors and residual PCs 1&5.

Notes. Red and black solid curves denote the sharpe ratios of Residual PC 1 and 5, respectively. The six dashed curves denote the sharpe ratios of the FF5+1 factors.

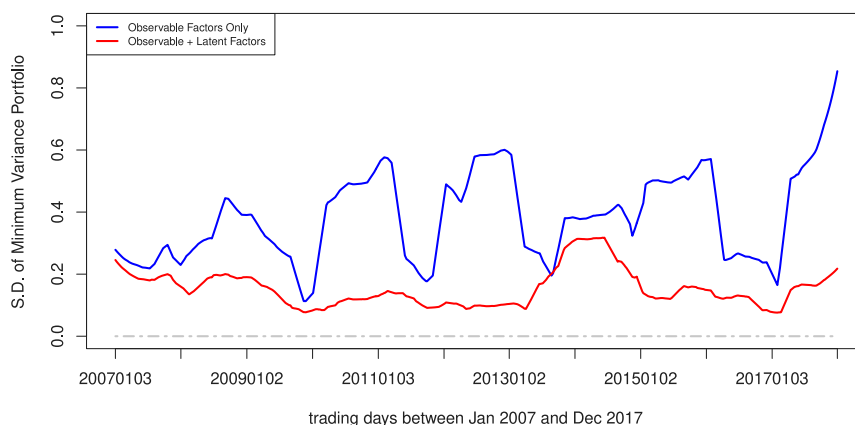


Fig. 8.5. Performance comparison in construction of minimum variance portfolio.

the proposed model outperforms the purely observable factor model. We have also identified new anomalies from the residual PCs and the unsupervised exogenous components.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jeconom.2025.106058>.

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